



Accenture Duck Creek

Duck Creek Billing™ Implementation Guide





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Duck Creek Technologies. <http://www.duckcreektech.com>

Duck Creek Billing™ Implementation Guide

The Duck Creek Billing™ Implementation Guide takes the billing user from installation to a customer implementation ready for the hands-on billing user. The guide is organized by feature, and includes the appropriate configuration, customization, and integration work needed to use each feature.

For more information about using billing, see the *Duck Creek Billing™ User Guide*. For more information about the features included in billing, see the *Duck Creek Billing™ Features Overview*.

Duck Creek Billing™ Installation

Duck Creek Billing™ requires installation of the following EXAMPLE Platform® products:

- EXAMPLE Server® must be installed first
- EXAMPLE TransACT®
- EXAMPLE Asynchronous Request Viewer
- EXAMPLE Forms™



Installation of EXAMPLE Express® is also recommended, and is required for the installation of the Billing On Demand Components. Without EXAMPLE Express®, Duck Creek Billing™ pages will not be viewable out of the box.

System requirements for Duck Creek Billing™ are:

- Microsoft Internet Explorer 8 or higher
- Microsoft Message Queuing
- Internet Information Service (IIS) v.7



When installing IIS for the first time, the uninstall .NET Framework must be uninstalled first. After successfully installing IIS, reload .NET Framework

- Microsoft .NET Framework v3.5 (SP 1)
- Microsoft SQL Server 2005 (SP 3) with Full Text Search (FTS) and Reporting Services enabled.



New SQL Server 2005 users should install FTS when installing Microsoft SQL Server 2005. Users who are upgrading should check in Control Panel>Administrative Tools>Services to see if FTS is installed and is started. If it is not shown in the Services folder, install it from the SQL 2005 CD.

The following services must be started to run Duck Creek Billing™:

- DuckCreek_MSMQBillingListener_EXAMPLE SERVER
- DuckCreek_Async_Request_Queue_EXAMPLE Async Queue O
- DuckCreek Scheduled Requests EXAMPLE Billing Components O
- Microsoft Excel 2003 for generating Excel format agent commission statements (EXAMPLE Platform® 4.2.2. or later)



Compatible versions of Microsoft Excel and SQL Server are required. SQL Server 2005 requires Microsoft Excel 2003.

Installing Duck Creek Billing™



When upgrading Duck Creek Billing™, be certain to upgrade all EXAMPLE Platform® products to the same version number to ensure that all the installed products work as expected.

To install Duck Creek Billing™:

Space availability and requirements for the selected features appear on the screen.



To install all the features and functionality available in Duck Creek Billing™, select both check boxes.

1. Download the Duck Creek Billing™ installation file from the *Duck Creek Technologies, Inc. Support Site*.
2. Execute the installation file.
The contents of the file are extracted, and the system prepares the InstallShield® Wizard. One of the following occurs:
 - **If this is not the first time you have installed Duck Creek Billing™** — The Existing Installed Instances Detected screen appears. Continue with step 3.
 - **If this is the first time you are installing Duck Creek Billing™** — The Welcome screen appears. Continue with step 5.
3. On the Existing Installed Instances Detected screen, select one of the following options:
 - **Install a new instance of this application** — If you select this option, continue with step 4.
 - **Maintain or update the instance of this application selected below** — If you select this option, follow the instructions in *Modifying or Repairing an Installation*.
4. Click **Next**.
The Welcome screen appears.
5. On the Welcome screen, click **Next**.
The License Agreement screen appears.
6. Read the License Agreement carefully, and then do one of the following:
 - Click **I accept the terms of the license agreement** to accept all terms of the agreement.
 - Click **I do not accept the terms of the license agreement**, and then click **Cancel** to end the installation process.
7. (Optional) If you wish to print the license agreement, click **Print**.
8. Click **Next**.
The Select a Feature screen appears.
9. Select the check boxes for the features you wish to install.
Options are:

- Billing Server Components (Requires previous instance of EXAMPLE Server®.)
 - Billing OnDemand Components (Requires previous instance of EXAMPLE Express®.)
10. Click **Next**.
The Instance Name screen appears.
 11. Enter a unique instance name for the Duck Creek Billing™ install, or accept the default.
 12. Click **Next**.
The system begins installation and opens the configuration portion of the Duck Creek Billing™ installation.
The Configure Database Credential screen appears.



Before beginning the configuration portion of an installation, the system checks to see if Microsoft Message Queuing is installed. If it is not installed, an alert appears asking if you wish to install MSMQ Messaging now. Clicking **Yes** aborts the configuration but does not install MSMQ Messaging. You must use Add/Remove Programs to install it manually, and then run the configuration portion again. If you click **No**, the installation process will continue normally, but Duck Creek Billing™ will not work properly until MSMQ Messaging is installed.

Configuring Database Credentials

When the Configure Database Credentials screen appears, all databases available for installation are displayed in the Configure window.

To configure database setup and information:

1. On the Configure Database Credentials screen, select the appropriate database from the list.
2. Select the credentials to use for database installation and maintenance.
Options are:

- **Windows authentication** — Use the current user's credentials.
- **SQL Server authentication** — Use the **Login ID** and **Password** entered.



To select **SQL Server authentication**, your SQL Server Installation must be configured to allow SQL Server authentication.

3. Click **Next**.
The Reporting Services screen appears.

Configuring Reporting Services

To use the reporting features of Duck Creek Billing™, Reporting Services must be turned on and configured.



If you select the **Skip reporting services configuration** check box, reporting features will not be available in Duck Creek Billing™.

To configure reporting services:

Options are:

1. On the Configure Reporting Services screen, select the credentials to use for Installation/Maintenance.
 - **Connect using your current credentials** — Connect to the report server as the user running the install.
 - **Connect using the EXAMPLE Server® credentials (DCTWP User)** — Connects to the report server as the DCTWP User or whatever is specified in the Worker Process login ID during database credential configuration.



At runtime the credentials for the DCTWP User are used.

2. In the **Report Server** field, enter the name of the server to use for the SQL Reporting Services web service
3. (Optional) Click **Test** to test the reporting server.
4. In the **Deploy reports to this report server folder** field, enter the name of the folder to store the reports in.
5. Click **Next**.

Executing the Duck Creek Billing™ Installation

When the installation is ready to configure your system for Duck Creek Billing™, the following list of installation items appears:

- XML Files
- Manuscripts
- Database Setup
- Reporting Setup

By default, the installation executes all items in this list. If you do not wish the installation to execute an item, clear the check box beside that item.



Clearing a check box causes the installation to *not* execute a process and may prevent Duck Creek Billing™ from running correctly.

During installation, if a database does not exist, the installation creates that database and executes all selected processes. If the database already exists, however, the installation compares the existing database structure to the appropriate scripts. (You can view a list of differences between the existing structure and these scripts by clicking the **Preview Changes** button.) If the **Database Setup** check box is selected, the installation then makes the appropriate changes to the existing database tables and columns.



To see changes to the **XML Files** and **Database Setup** items, click **Preview Changes** beside the appropriate item.

To execute the selected installation items:

1. Click **Finish**.

As the installation executes each process, that process becomes shaded.

2. Upon completion, the system displays a message indicating whether configuration was successful. Click **OK**, and then do one of the following:
 - If the configuration was not successful, a **View Errors** button appears. Click **View Errors** to open a text file with the installation errors listed. When finished, click **Close**, and then continue with step 4.
 - If the configuration was successful, click **Close**.
3. (Optional) If necessary, click **Back** to return to the previous window and select or clear any check boxes.
4. If the installation opens another InstallShield Wizard dialog box, click **Finish**.

Configuring Services for Duck Creek Billing™

Services configuration files are located at: C:\Program Files\Duck Creek Technologies\Shared\Bin\DuckCreek.Utility. To edit these files, open them in a text editor program.

The following services must be configured:

- MSMQBillingListenerSvc.exe.config
- ScheduledRequestSvc.exe.config

MSMQBillingListenerSvc.exe.config

The Microsoft Message Queue (MSMQ) Billing Listener is a message store that enables messages to be read or written. When a message is read, it is removed from the queue. The MSMQ is accessible via an API.

Key	Definition	Default
ServerConfig	The location of the Server.config file.	C:\Program Files\Duck Creek Technologies\Shared\Server.config
Request Url	The URL of the EXAMPLE Server® that the service will post requests to.	<i>http://localhost:8080/DuckCreek/DCTServer.aspx</i>
DuckCreekUserID	The user ID.	admin
DuckCreekPassword	The user password.	admin
ErrorLog	Specifies the name of the Event Log to log errors for this service.	DCTMSMQBillingListener
OverwriteErrorLog	This setting forces old event entries to be overwritten as the Event Log fills up.	1 (True)



When moving from testing to production, it is recommended the message queue be cleared and the DuckCreek_MSMQBillingListener_EXAMPLE SERVER be restarted to ensure that only valid production messages are in the queue.

ScheduledRequestSvc.exe.config

The ScheduledRequestSvc specifies the frequency to check for scheduled requests.

Key	Definition	Default
ServerConfig	The location of the Server.config file.	C:\Program Files\Duck Creek Technologies\Shared\Server.config
RequestIntervalInMilliseconds	Specifies the time in milliseconds between checking for requests.	60000 (one minute)
DuckCreekUserID	The user ID.	admin
DuckCreekPassword	The user password.	admin
RequestUrl	URL of the EXAMPLE Server® responsible for working with the specified web site to ensure the correct application pool is recycled.	http://localhost:8080/DuckCreek/DCTServer.aspx
AppLogSource	The source description used by the service when logging an error to the Event Log.	DCTSchedRequestSvc
AppLogName	The name of the Event Log to log errors for this service.	Application
RequestFile	The XML containing the request.	ScheduledRequests.xml
RequestToCheckForSuccess	The name of the request to check for successful activity scheduling.	ScheduledActivity.queueListRq
RequestTimeoutInMilliseconds	The amount of time in milliseconds the service waits for EXAMPLE Server® to respond when sending a request to the URL.	120000 (two minutes)

The User Interface

Duck Creek Billing™ is for real-world people performing real-world tasks. The intuitive user interface offers quick, functional access to the information and processes that users access most frequently. Key elements of the user interface include:

- The Billing Dashboard
- The Account Home page
- The Policy Term Summary page
- Controls (grids, dates, and validation)



Duck Creek Billing is a highly customizable and configurable product. Not all pages may appear as described in this user guide, which is based on out-of-the-box settings.

The Billing Dashboard

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Accenture Duck Creek Billing

Policy Billing Party Logout

Account Number

Billing Dashboard Billing Administration Batch Cash Management Tasks Notifications

Billing Dashboard

Welcome to the Billing Dashboard, Express Admin.

Activities

[Search For A Payment](#)

[Enter A Single Payment](#)

[Unidentified Payments](#)

[Create A New Account](#)

[Import Statements](#)

Accounts Last Accessed

Reports

[Aged Receivables](#)

[Cash In Suspense](#)

[Commission Activity](#)

[Daily Cash Journal](#)

[Disbursement Activity](#)

[Disbursements Pending Authorization](#)

[General Ledger](#)

[Payment Batch](#)

[Transaction Balancing](#)

[Unidentified Cash](#)

Charts

The current system date is May 16, 2012

Find a Billing Account

I want to find an account by searching for

Search

Notifications

You have no notifications.

Tasks

You have no tasks due today.

[Change Password](#)

From the Billing Dashboard page, a user can:

- Search for a payment.
- Enter a single payment from the Cash Management screen instead of within an account.
- View summary information for payments automatically or manually entered that were not posted to a policy or account.
- Create an account in the billing system without going through policy issuance.
- View a list of recently created or accessed accounts.
- Open an available report with one click.
- View notifications and tasks.

The Account Home Page

The Account Home page is the primary workhorse for the Duck Creek Billing™ system, and provides a direct path for accessing and managing billing activities at the account level.

Account Home (Add A Note)

Most Recent Invoice

Invoice Date	Docs	Amount	Due Date
5/16/2012		\$7.32	6/5/2012

Last Payment Received

Date	Amount	Method
5/16/2012	\$7.32	Check

Next Scheduled Invoice

Invoice Date	Amount	Due Date
6/15/2012	\$7.32	7/5/2012

Recent Account Activity

Date	Activity Type	Created By	Amount
5/16/2012, 1:48 pm	Payment Received	Express Admin	\$7.32
Check payment for \$7.32 received. Check Number: Receipt date: 05/16/2012			
5/16/2012, 1:44 pm	Invoice	Express Admin	\$7.32
Invoice created. Total amount due \$7.32. Due date 06/05/2012			
4/14/2012, 1:19 pm	Payment Received	Express Admin	\$7.32
Check payment for \$7.32 received. Check Number: Receipt date: 04/14/2012			
4/14/2012, 1:17 pm	Invoice	Express Admin	\$7.32
Invoice created. Total amount due \$7.32. Due date 05/05/2012			
3/16/2012, 1:17 pm	Payment Received	Express Admin	\$7.32
Check payment for \$7.32 received. Check Number: Receipt date: 03/16/2012			
3/16/2012, 1:16 pm	Invoice	Express Admin	\$7.32
Invoice created. Total amount due \$7.32. Due date 04/05/2012			
3/5/2012, 1:12 pm	Policy Transaction	Express Admin	\$64.05
New Business: receivable has been processed.			
3/5/2012, 1:12 pm	Policy Term Issued	Express Admin	N/A
Policy Term 000001.03/05/2012 - 03/05/2013 created			
3/5/2012, 1:12 pm	Account Created	Express Admin	N/A
Account created.			

9 results found. Currently showing 1 - 9.

The Account Home page provides quick access to view and work with:

- The most recent invoice.
- The account history, sortable by date, activity type, or policy term in the Recent Account Activity list.
- A list of all policy terms associated with the account, with a link for quick access to the Policy Term Summary page for each term.
- A summary of the account ledger and account details, providing a quick look at the account status.

- ## The Policy Term Summary Page

Account Number
[Policy](#)
[Billing](#)
[Party](#)
[Logout](#)

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[Billing Dashboard](#) |
 [Billing Administration](#) |
 [Batch Cash Management](#) |
 [Tasks](#) |
 [Notifications](#)

test one
000001
(Active)

[Account Home](#)

[Instalment Schedule](#)

[Account Settings](#)

[Make A Payment](#)

[Items In Suspende](#)

[Account Tasks](#)

Policy Term Summary
Add A Note

Summary			
Policy Number	Eff/exp Dates	Status	Aggregation Reference
000001	3/5/2012 - 3/5/2013	Active	

Most Recent Invoice

Invoice Date	Docs	Amount	Due Date
5/16/2012		\$7.32	6/5/2012

Next Instalment

Date	Amount	Due Date
5/16/2012	\$0.00	6/5/2012

Future Policy Term Renewal

Account Number	Policy Term Settings

Additional Data

Name	Value				
Policy Term Activity					
Date	Transaction	Receivable	Amount	Bill In Full	CommissionAmt
3/5/2012, 12:00 am	New Business	Premium	\$61.00		\$0.00
Bill Item Effective Dates - 3/5/2012 - 3/5/2013					
3/5/2012, 12:00 am	New Business	Tax	\$3.05		\$0.00
Bill Item Effective Dates - 3/5/2012 - 3/5/2013					
Other Policy Terms Associated with this Policy					
Policy Term	Policy Status	Current Due	Outstanding Balance		
Policy Term: 3/5/2012 - 3/5/2013 (1 Item)					
3/5/2012 - 3/5/2013	Active	\$0.00	\$42.72		

The current system date is May 16, 2012

Policy Term Actions

[Transfer Policy Term](#)

[View/Edit Policy Term Settings](#)

[Hold Billing for Policy Term](#)

Policy Term Details

Billing Frequency: MONTHLY

Master Company: DCT

Line of Business: LNI1

Equity Date: 7/5/2012

Policy Term Ledger

Outstanding Balance:	\$42.72
Current Amount Due:	\$0.00
Past-Due Amount:	\$0.00
Receivable Amount:	\$64.05
Paid on Policy Term:	\$21.33
Written Off/waived:	\$0.00

Agent

Insured

This page provides answers to the most frequently asked questions concerning policy account terms. All tasks related to policy terms are quickly launched from this page.

From the Policy Term Summary page, the user can:

- Transfer a policy term to another policy.
- View or edit policy term settings.
- Update renewal information.
- Hold billing activity for the policy term.
- View policy term details.
- View detailed information for the insured and the agent.
- Access a list of other policy terms associated with the account.

Working with Controls

The Duck Creek Billing™ user interface incorporates ExtJs2.2, making available a diverse set of GUI-based form controls (widgets) including:

- Grids and filters
- Date fields
- Validation functions

Grids and Filters

In Duck Creek Billing™, users can change the items displayed in a grid using filters or special options on the grid. Items that are displayed in a list may be determined by using the Support_CodeListExclusion database table. If the user selects an item from the grid that is not displayed, that selection will override the exclusions, and the item will appear on the page. See *Working with Grids and Filters* in the *Duck Creek Billing™ User Guide*.

Grids in Duck Creek Billing™ integrate a wide variety of tools that allow users to easily view customized information. These grid types include:

- **Grid Types**
 - **Grouping grids** – Group items within the grid. Common rows are grouped together based on the grid's configuration.
 - **Preview grids** – Display an extra information row per item. This information row can be collapsed or expanded (for a single row or for all rows) using a button in the grid toolbar.
- **Check box grid** – Display a check box on each row, based on a row rule check. These grids can store all selected rows in a paging grid, storing the items selected on each page until the user is ready to submit the selections.
- **Grid toolbar** – Allow users to display or hide items according to their preferences.
- **Column filtering** – Allows users to filter grid data within each column. Data is displayed based on all filters the user has selected. Filtering types include:
 - **Dates** – Allows **less than or equal to**, **greater than or equal to**, and **equal** filtering.
 - **Text** – Allows text searches.
 - **Amounts** – Allows **less than or equal to**, **greater than or equal to**, and **equal** filtering.
 - **Support table values** – Loads selectable items using check boxes.

- **Row classification** – Allows a class name to be appended to a row to change styling. This classification is based on a rule provided through a JavaScript function.
- **DataOnLoad event** – Allows the onload event to occur immediately after a page or image is loaded.

Date Fields

Duck Creek Billing™ offers the following date field functionality for date fields:

- System pages with their own Ext Js date fields that do not interfere with interview pages.
- Available events for date fields:
 - onBlur event
 - onChange event
 - onKeyUp event
 - onKeyPress event



Date fields are turned on by providing a function that is called for one of the listed events. The function is passed as a configuration parameter in the XSL. If the screen XSL does not pass a function, the event is not processed.

Validation Functions

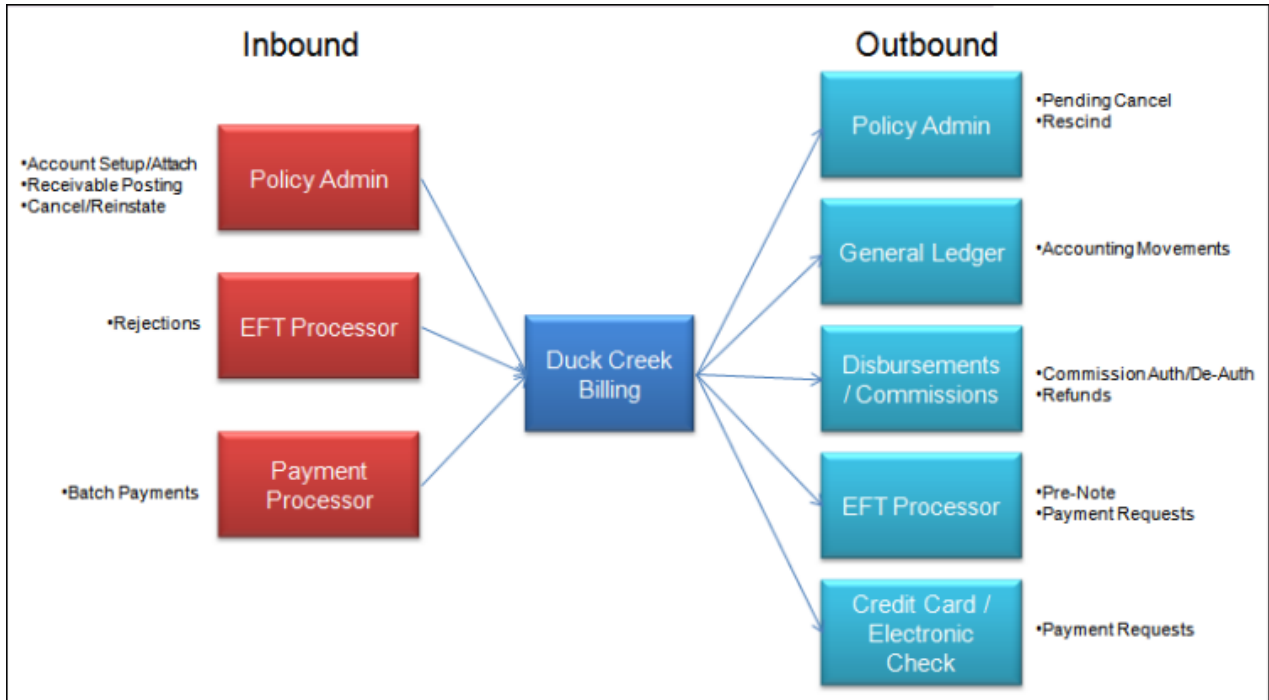
- **Credit Card Validation** – The credit card validation function performs a behind-the-scenes validation on all entered credit card numbers. All standard credit card types are handled.
- **Time Validation** – The Time Validation function checks that any entered time is valid.

Integrating Duck Creek Billing

Third-party products may be integrated with Duck Creek Billing™ to complete a billing package. Easy to configure integration points allow the user to integrate Duck Creek Billing with payment processing for EFT accounts, lockbox payments, and policy administration systems.

Integrating with Inbound and Outbound Processes

Duck Creek Billing™ integrates with a number of other systems, as illustrated below. Processes can be inbound, outbound or bi-directional.

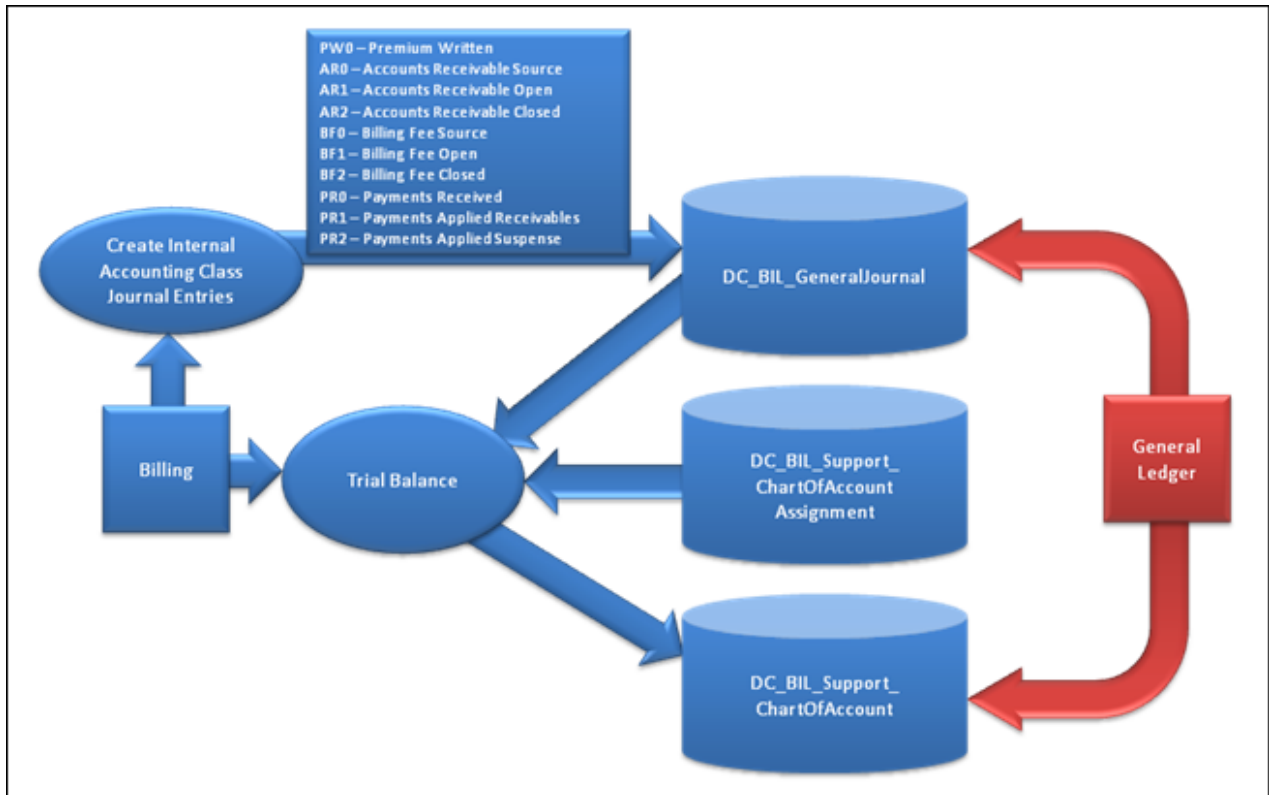


Systems integrated with Duck Creek Billing™ include:

- Policy administration systems
- General Ledger
- EFT Payments
- External Payment Processing
- Third Party Payment Interface
- Disbursement Interface

Integrating with the General Ledger

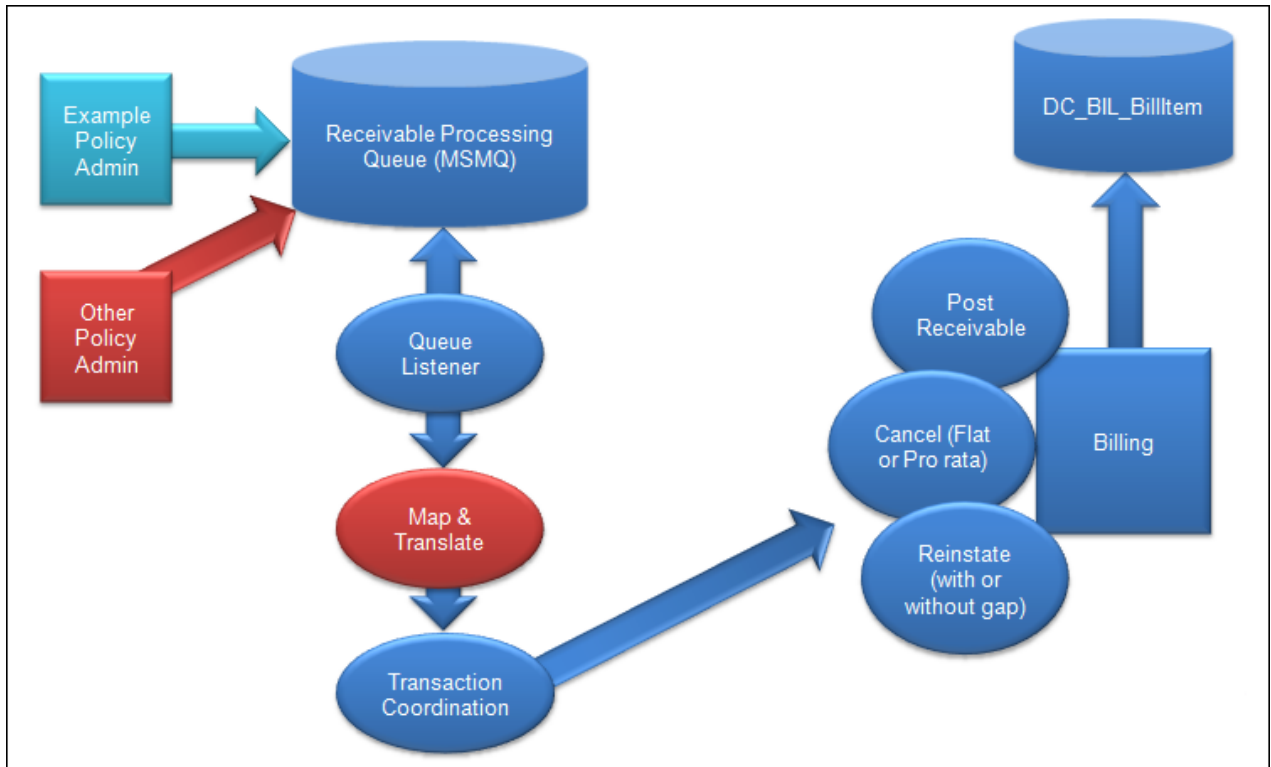
The integration process between Duck Creek Billing™ and a general ledger system is illustrated below.



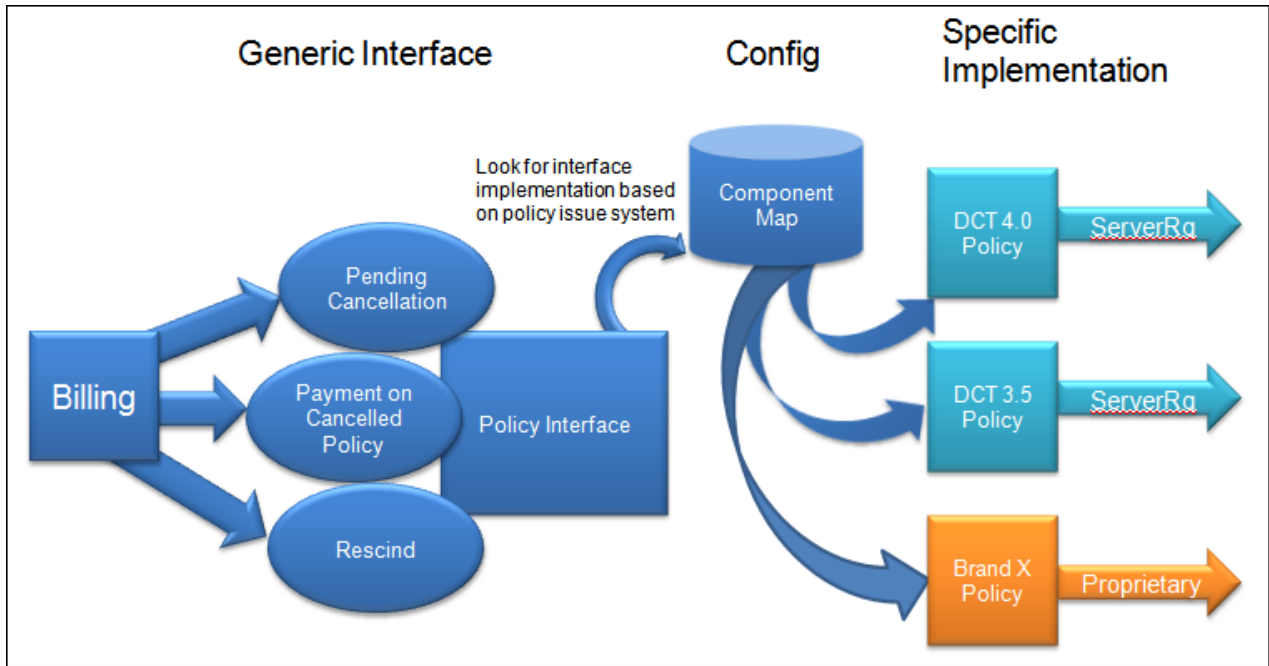
General Ledger systems can be interfaced with Duck Creek Billing™ using database tables. For more information about this feature, contact your Duck Creek Account Manager.

Integrating with Policy Administration Systems

The integration between Duck Creek Billing™ and policy administration systems is bi-directional. Data flows from the policy administration system to Duck Creek Billing™, as illustrated below.



The integration process from Duck Creek Billing™ back to the policy administration system is illustrated below.



The following interfaces may require customization to support a specific Duck Creek Billing™ implementation.

- Policy to Billing
 - Account, Policy Term, and Party Setup
 - Receivable Posting
- Billing to Policy
 - Pending Cancel
 - Rescind Pending Cancellation
 - Payment on Canceled Policy

Out of the box, Duck Creek Billing™ is integrated with EXAMPLE Express®. It is only necessary to hook up individual product lines to the Duck Creek Billing™ system.

To integrate Duck Creek Billing™ with a Product Line:

1. Using the EXAMPLE Author® import tool, import the Billing Policy Integration Framework Manuscript (all fields) into the product line.



These fields do not exist inside current Policy Manuscripts, including any Commercial Lines Templates 5.x.x.x and earlier releases. These fields can be added by importing the fields from the Manuscript Billing_PolicyIntegration_Framework_1_0_0_0.xml, which is included with the Duck Creek Billing™ install specifically for this purpose.

2. In the Policy Pages Manuscript, on the Submission page, create a **Book** button, using the express action **billingStart:billingStart**.

3. To display the new Account ReferenceID after the **Book** button has processed, set the Submission page to show the value of the text `_AddAccountResponse.value` field.
4. (Optional) To generate desired feedback data, connect the `BillingKeysPrivate.SetData` field to execute both prior to the Book express action and after the return response.



Suggestion: Connect `BillingKeysPrivate.SetData` to the `data.TotalResult` field

To integrate Duck Creek Billing™ with external policy administration systems:

1. The `_BillingInputDoc` must be set inside the working session on the Billing server. This data can be populated in multiple ways. Due to changes and customization, the carrier should determine paths and data of this session document by referring to the Account and Policy Plan manuscripts. These manuscripts can also set the data if desired, by calling the `<ManuScript.initDataRq>` request.
2. Normal billing workflow or any modification implemented by the customer can be carried out.
3. Once this data is set, the external system can call the Billing server with the **`billingStart:billingStart`** express action, and a `returnURL`. Once billing is done, the `_BillingOutputDoc` is available in the billing session for retrieval using the request API.

For more information, contact your Account Manager.

Creating Accounts and Users

The following section includes information covering the types of accounts that can be created, and the types of accounts that are necessary to use specific billing features including Monthly Premium Reporting and the Agency Lite feature.

End-User Functionality and Role-Based Security

Out-of-the-box roles are provided as a framework for customization of the role-based security features implemented in Duck Creek Billing™. Roles can easily be customized to allow users to have specific privileges not available with the out-of-the-box role. Shipped roles include: administrator, agent, senior customer service representative, and standard customer service representative.

Using the available roles and privileges, the following representative users can perform a variety of tasks:

- Policy system user
- Customer service representative
- Payment processor
- Billing administrator

Working with Privileges and Out-of-the-Box Roles

Duck Creek Billing™ provides four out-of-the-box roles that include specific privileges for an administrator, agent, senior customer service representative or standard customer service representative. These roles may be used as shipped to provide a user with a specific set of privileges, or privileges may be added or removed from the out-of-the-box roles. For example, using the out-of-the-box senior customer service representative role as a foundation, additional privileges may be added to allow the user to complete work not specified in the out-of-the-box role. The roles and associated privileges are managed in EXAMPLE UserAdmin™. The range of activities any given individual may

perform within Duck Creek Billing™ is determined through the roles granted to that person using EXAMPLE UserAdmin™.



The BillingLogin Role allows the user to login to Duck Creek Billing™, and is required for all users other than those with system administration privileges.

Privilege	Out-of-the-Box Roles			
	BillingAdmin	BillingAgent	BillingSeniorCSR	BillingStandardCSR
Access Billing Account Summary page		x	x	x
Access Account Summary page for Commission Accounts				
Access Disbursement Detail page			x	x
Access to the Installment Schedule page		x	x	x
Items in Suspense page			x	x
Payment Detail page			x	x
Policy Summary page		x	x	x
Receivable Write off/Waive Detail page			x	x
Unidentified Payment page and assign payments to an account			x	
Allows the user to do anything in billing	x			
Batch cash management facilities			x	
Add notes to an account		x	x	x
Authorize disbursements	x			
Hold billing and follow up activities			x	x
Update the target due date of an account			x	x
Change payor or paynee on an account		x	x	x
Change policy plan settings			x	x
Change policy plan settings for a renewal term			x	x
Change the hold type on an item in suspense			x	x

Allows user to change the system date				
Create an account without a policy			x	x
Schedule disbursement service interface requests		x		
Edit scheduled EFT requests				
View the Error Queue				
Hold Billing and Follow up Activity				
Enter a payment inside an account			x	x
Enter a payment outside an account, including unidentified payments			x	x
Manage Hold Events			x	
Manually redistribute installments		x	x	x
Manually allocate money in suspense			x	x
Manually rescind a policy in PCN status			x	
Manually re-suspend a previously disbursed disbursement				
View, create, edit, delete, and process payment batches			x	x
Search for a payment			x	x
Re-bill disbursements			x	
Manually override the refund to source check				
Reprint an invoice			x	x
View the billing aged receivables report				
View all billing reports				
View billing cash in suspense report				
View billing commission activity report				
View billing daily cash journal report				
View Disbursement Activity report				

View billing disbursements pending authorization report				
View billing general ledger report				
View Billing payment batch report				
View transactional balancing report				
View unidentified cash report				
Request a disbursement up to a specified limit			x	
Resuspend a pending disbursement			x	x
Reverse a payment			x	
Submit monthly payroll reports			x	x
Suspend an allocated payment			x	
Transfer cash from suspense			x	x
Transfer a policy between accounts			x	
Transfer a policy term to a collections account			x	
View image associated with an invoice		x	x	x
Access reports			x	x
Write off cash in suspense			x	
Write off or waive receivables			x	x

Selecting Billing Account Options

Billing plans may be configured by specifying the values for Billing Account Options. The available options control all automated billing and collection processes, including account type, installment scheduling and invoicing and collection follow-up activities. The options Duck Creek Billing™ users see is determined by the ManuScript configuration.



Any modifications to existing Billing Account Option configurations will affect all billing accounts and policies which reference the configuration.

The following table includes all the possible options, and may include options that are not configured in your selected ManuScript.

Category	Option	Definition	Valid Entries	Default
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Payment				
	Account Type Code	Specifies the type of account.	• Account • Agency Lite	Account
	NSF Charge Action	Specifies whether a charge is assessed for non-sufficient funds (NSF).	• A — Assess NSF fee. • N — Do not assess NSF fee.	A
	NSF Reversal Charge Amount	Amount of the NSF penalty.	Any value, including 0.	\$25
	Payment Reversal Due Date Adjust Reason	When a payment is reversed, the system re-opens the bill items that were closed by the reversed payment. The system adjusts the due date for the closed items only when a payment reversal reason is indicated here.	• CM — Company mistake. • NSF — Non-sufficient funds.	CM
	Excess Payment On Policy	Defines how excess payment is handled when a payment allocation is directed to a specific policy with a zero balance.	• A — Payment will be automatically allocated to other policies on the account that have a balance. • S — Suspend the payment on the policy it was intended for, and allow for disbursement.	A

Scheduling				
	Planned Installment Frequency Code	The frequency at which billing occurs on the account level.	Monthly	Monthly
	Bill Production Offset Days	The number of days between the invoice date and the due date. Determines when the invoice is generated.	Any positive number between 1 and 31.	20
	EFT Draw Offset Days	The number of days between EFT invoice date and the due date. This value is added to the EFT invoice date to determine the EFT draw date.	Any positive number between 1 and 31.	3
	Out of Cycle Bill Allowed Indicator	Determines whether an out of cycle bill is allowed.	Select check box to allow.	Not selected
	EFT Pre Note Indicator	Determines whether the account sends pre-notes when it is first set up on EFT.	Select check box to allow.	Not selected
	Default Policy Plan Manuscript ID	Specifies the ID of Manuscript being used.	Any valid Manuscript ID.	Billing_PolicyPlanSetup_TemplateFull_1_0_0_0
Disbursement				
	Disbursement Method Code	Determines the method by which disbursements will be sent to the payee on the account.	• Check • EFT • Credit Card	Check
	Hold Disbursement Indicator	Determines whether the account allows for automatic disbursement requests.	Select check box to allow.	Not selected

Invoice				
	Account Rollup	Displays the total amount due on the invoice.	Select check box to allow.	Selected
	Aggregation Rollup	Displays the aggregation on the invoice.	Select check box to allow.	Not selected
	Bill Item Rollup	Displays bill items on the invoice.	Select check box to allow.	Not selected
	Commission Rollup	Displays commission on invoice.	Select check box to allow.	Not selected
	Coverage Rollup	Displays coverage on invoice.	Select check box to allow.	Not selected
	Form MS	Specifies forms ManuScript.	Any valid ManuScript.	BillingInvoice
	Line Type Rollup	Displays line type on invoice.  If this check box is not selected, no information will display on the invoice even if other check boxes are selected.	Select check box to allow.	Selected
	Policy Rollup	Displays the amount due on each policy below the Premium heading on the invoice.	Select check box to allow.	Selected

	Receivable Sub Type Rollup	Displays the receivable sub type on the invoice.	Select check box to allow.	Not selected
	Receivable Type Rollup	Displays each receivable on the invoice.	Select check box to allow.	Selected
	Transaction Type Rollup	Displays the transaction type in the description box on the invoice.	Select check box to allow.	Not selected
	Unit Rollup	Displays the unit on the invoice.	Select check box to allow.	Not selected
	Invoice Method	Specifies the invoice type.	• Paper • EFT	Paper
	EFT Service Charge Amount	Specifies the service charge amount for EFT invoice.	Any value.	\$1
	EFT Service Charge Action	Determines whether the account will generate service charges for EFT invoices.	• Fixed Service Charge • No Service Charge • Variable Service Charge	Fixed Service Charge
	Invoice Service Charge	Specifies the service charge amount.	Any value.	\$1
	Paper Invoice Service Charge Action	Determines whether the account will generate a service charge for paper invoices.	• Fixed Service Charge • No Service Charge • Variable Service Charge	A

	Suppress First	Determines if the first invoice should be suppressed.	Select check box to allow.	Not selected
	Paper Invoice Service Charge Percent	Specifies the service charge percentage.	Any value.	8.00 (8%)
	EFT Invoice Service Charge Percent	Specifies the service charge percentage.	Any value.	8.00 (8%)
	Service Charge Minimum Amount	Specifies the minimum service charge amount assessed.	Any amount.	\$5.00
	Service Charge Maximum Amount	Specifies the maximum service charge amount assessed.	Any amount.	\$20.00
	EFT Variable Service Charge Exclusions	Specifies items excluded from the variable service charge calculation.	Any valid selection.	None
	First Name (EFT Only)	Specifies the first name on bank account used for the EFT draw.	Any valid entry.	None
	Last Name (EFT Only)	Specifies the last name on bank account used for the EFT draw.	Specifies the last name on bank account used for the EFT draw.	None
	Bank Name (EFT Only)	Bank Name (EFT Only)	Any valid entry.	None
	Account Type (EFT Only)	Specifies the account type.	Checking Savings	None
	Routing Number (EFT Only)	Specifies the bank routing number.	Any valid entry.	None

	Account Number (EFT Only)	Specifies the bank account number.	Any valid entry.	None
	Target Draw Days (EFT Only)	Determines the day of the month the EFT draft is drawn from the specified account.	Any positive number between 1 and 31.	1
	Paper Variable Service Charge Exclusions	Specifies items excluded from the variable service charge calculation.	Any valid selection.	None
	Invoice Line Type (Paper Only)	Specifies the items to exclude from the variable service charge calculation.	(None) Earned Premium First Invoiced Premium Missed Invoice Premium Prior Invoiced Premium Past Due Premium First Invoiced Charges Prior Invoiced Charges Past Due Charges (All)	None
	Receivable Type (Paper Only)	Specifies the receivable type for the items to exclude from the variable service charge calculation.	(None) Premium Tax Fee Surcharge ERPM Deposit (All)	Deposit

	Receivable Sub Type (Paper Only)	Specifies the receivable sub type items to exclude from the variable service charge calculation.	(None) PREV DBRB N L S (All)	(All)
	Transaction Type (Paper Only)	Specifies the transaction type for the items to exclude from the variable service charge calculation.	(None) Endorsement Flat Cancellation New Business Pro Rata Cancellation Monthly Report (All)	(All)
	Use Projected Service Charge	Determines if the projected service charge is displayed on the Installment Schedule.	Select the check box to utilize.	Selected
	Target Due Days	Determines the day of the month that invoices for the account are due.	Any positive number between 1 and 31.	1
Suspense				
	Disbursement Maximum Amount	Specifies the suspended amount that can be disbursed automatically at the time the Disbursement Write Hold Days is reached. If the amount is more than the Disbursement Maximum Amount, the money is held in suspense.	Any positive value.	\$99,999.99

	Disbursement Minimum Amount	Specifies the suspended amount that can be disbursed automatically at the time the Disbursement Write Off Hold Days is reached. If the amount is less than the Disbursement Minimum Amount, the money is written off.	Any positive value.	\$10
	Disbursement Write Off Hold Days	Specifies the number of days money will be held in suspense before it is evaluated for disbursement or write off.	Any value including 0.	30
	Cash With Application Disbursement Write Off Hold Days	Specifies the number of days payments marked with "Cash with App" payment type will be held in suspense before evaluation for disbursement or write off.	Any value including 0.	30
Follow Up				
	Late Charge Action	Determines whether the account will be charged late fees.	• Charge • None	Charge

	Late Charge Amount	Determines the amount of the late charge. Valid only when the value of the Late Charge Action setting is Charge. Late charges appear on the next scheduled invoice, and are not included in the current amount due.	Any value.	\$10
	Late Charge Days	Determines the number of days past the due date the late charge is assessed if the unpaid invoice amount meets or exceeds the Late Charge Threshold Amount. Valid only when the Late Charge Action setting is C.	Any value.	3

	Late Charge Threshold Amount	Determines the amount of the invoice that must remain unpaid in order to assess the late charge on an account. Any amount less than the threshold amount does not generate a late charge, and is written off or billed with the next scheduled invoice.  Valid only when the Late Charge Action setting is Charge.	Any value.	\$10
	Reminder Notice Action	Determines whether reminder notices are sent on the account.	• Reminder • None	Reminder
	Reminder Notice Days	Determines the number of days before or after the due date that the reminder notice can be generated if the unpaid invoice amount meets or exceeds the Reminder Notice Threshold Amount. The value of the Reminder Notice Action Setting must be set to Reminder.	Any value.	5

	Reminder Notice Threshold Amount	Determines the unpaid amount of the invoice that must remain unpaid to generate a reminder notice for an account. Any amount less than the threshold amount will not generate a reminder notice, and will be written off or billed with the next scheduled invoice. The value of the Reminder Notice Action setting must be Reminder.	Any value.	\$10
	EFT Default Letter Manuscript ID	Specifies the Manuscript to be used.	Any valid Manuscript ID.	
	EFTPCN Evaluation Days	Specifies the number of calendar days after an EFT invoice due date that a policy term will be evaluated for pending cancellation status.	Any positive integer.	0
Commissions				
	Commission Statement Format	Specifies commission statement format.	• Adobe PDF • Excel • Comma Separated Value (CSV) • XML	Adobe PDF

	Commission Statement Frequency	Specifies commission statement frequency and day of the month for processing.	- Monthly - Any number between 1 and 31.	- Monthly 31
	Commission Statement Payment/Collection Method	Specifies the method for paying and collecting commissions.	- Check - EFT	Check

Selecting Policy Plan Options

Policy plan options may be specified by determining the values for policy plan options. At the policy-level, these options control invoice frequency, commission scheme, follow up for pending cancel, payment allocation and down payments.

The table below specifies what policy options are available for configuration. ManuScript configuration determines which options are available to Duck Creek Billing™ users.

The following table includes the policy options available.

Category	Option	Definition	Valid Entries	Default
Scheduling				
	Frequency	Determines the planned installment frequency code for invoicing.	- Monthly - Quarterly - Annually - Semi-Annually - Bi-Monthly	Monthly
	Installment Plan Description	Specifies a free-form description of the installment plan.	- None - One less than term - Two less than term	One less than term
	Late Entered Strategy Code	Determines whether to catch up or spread the premium when a bill item is posted to billing after its effective date.	- Catch up - Spread - None	None
	Max Number of Installments	Determines the maximum number of installments for a policy term based on frequency.	12 invoices	12

	New Business Earliest Bill Offset Days	Determines the maximum number of days from the effective date of the new business policy term that the first installment is due.  Changing this setting after policy issuance will cause rescheduling activity to occur.	Any positive number, including 0.	45
	Last Available Due Date Displacement	Determines the minimum number of days prior to the expiration date of a policy term that an invoice can be due.  Changing this setting after policy issuance will cause rescheduling activity to occur.	Any positive number, including 0.	30
	Renewal Business Earliest Bill Offset	Determines the maximum number of days from the effective date of the renewal policy term that the first installment can be due.	Any positive number, including 0.	20
	Rescind Redistribution Strategy	Determines what to do with installments that did not invoice due to a rescission because the policy was in a pending cancellation status.	• Catchup • Spread	Catchup
	Reinstatement Charge Amount	Determines the policy term charge created in billing that is assessed when a policy term is reinstated.	Any positive value.	\$5
Commission				

	Commission Scheme	Determines how commission is paid on a policy.	• POW1 (Paid on Written) — The entire commission amount is authorized when the receivable is posted. • POP1 (Paid on Paid) — The commission is authorized as payments are applied to the receivable. • POP2 (Paid on First Paid) — The entire commission amount is paid when the first payment is applied to the receivable. • POE1 (Paid on Effective) — Commission is authorized for each policy transaction at issuance. Payment is not required. • POP4 (Paid on First Paid, plus latter of effective or activity date) — Receiving one good payment on the policy term releases the commission for the policy term. The latter of the policy transaction effective date or the activity date (process date) of the payment is used as the commission authorization date.	POW1
Follow Up				
	P C N Allow	Determines whether or not the policy term should ever be evaluated for pending cancellation.	• Selected — Yes • Cleared — No	Selected
	P C N Trig Days	Determines the number of days after the due date that a policy term is evaluated for pending cancellation.	Any positive integer.	5

	Auto Rescind Tolerance Days	Determines the tolerance number of days within the Auto Rescind Equity Date that a recalculated current equity date must be in order to rescind a policy term.	Any positive integer.	5
	Auto Reinstatement Days	Determines the number of days from the term expiration date that the equity term expiration date and the equity date must meet after a payment has been made on the cancelled policy. If the payment made on the cancelled policy term brings the equity date to be greater than the term expiration date minus the Auto Rescind Days, the system will send a request for reinstatement to the issue system.	Any integer between 0 and 365.	0

Changing Bill Plans at Policy Renewal or Re-issue

Out of the box, Duck Creek Billing™ does not allow users to change the selected bill plans at the account or policy level when a policy is renewed or re-issued. Customization is required to allow a bill plan to be changed at policy renewal or re-issue.

To change bill plans during policy re-issuance or renewal:



Code changes are required to retrieve the correct bill plan during the process of rendering the Renewal and Re-issue pages.

1. Verify that the Submission page displays the current bill plan for the policy being re-issued or renewed.
2. Verify that the options displayed in the **Select Billing Plan** list are compatible with the selected account level bill plan.
3. Open the **NewPolicyTermCommon.xml** file in Notepad or another text editor program.
By default this file is located at C:\Program Files\Duck Creek Technologies\Transform\.
4. Navigate to the **BillClass** section, and delete the **<!--** opening tag and the **-->** closing tag for the section to execute the code.
5. Save the file.
A different bill plan may now be specified when the policy is renewed or re-issued.

Working with the Agency Lite Feature

Direct bill systems typically use a gross premium basis for statement reconciliation. The Agency Lite feature in Duck Creek Billing™ allows agencies with low volumes of small agent accounts to reconcile statements on a net settlement of accounts basis using direct bill features.



Because manual reconciliation is required for each statement and for the creation of commission records, agency lite accounts are not recommended for large numbers of agents or for agents with more than 50 policies per statement.

Important considerations for agency lite accounts include:

- Net payments indicate that the agent is retaining the commission.
- Gross payments indicate that the commission should be paid to the agent.
- Manual data entry is not required if the statement is paid in full.
- Once an account setting is specified as gross or net, it cannot be changed.
- Net and gross commission payments cannot be mixed within an account.

Commission authorization records in direct bill are usually written to the CommissionAuthorization database table based on rules in the commission scheme specified in the Policy Plan Setup Options. For agency lite accounts, no commission scheme is specified, so no commission records are written.

The balances in direct bill always represent gross amounts. If payment is made in the net amount, an open balance representing the commission the agent has retained will remain.

Configuring Agency Lite Accounts

To configure the Agency Lite option:

1. In the **Billing Account Options** settings in the **Account Type** section, make the following selections:
 - In the **Account Type Code** list, click **Agency Lite**.
 - In the **Gross/Net Indicator** list, click the appropriate commission option.
2. Click **Save Changes**.
3. In the **Policy Plan Settings**, in the **Follow Up** section, clear the **PCNAllow** check box to prevent the Agency Lite policies from going into PCN status.
4. Click **Save Changes**.
5. Open the appropriate Manuscript.
6. In the **Commission_PolicyPlan.Scheme** field in the policy plan Manuscript, clear the default value to prevent commission authorization records from being written.

For information about entering payments for an Agency Lite account or for writing off commission adjustments, see *Working with Agency Lite Accounts* in the *Duck Creek Billing™ User Guide*.

Working with Monthly Reporting

Monthly Premium Reporting allows a business to pay higher premiums during the time the business is more active, and lower premiums during times of reduced activity. Pricing for an MPR policy is based on the annual estimated exposure or payroll subject to the rates, the exposure modification, and other factors the carrier has in the filing. By reporting the actual exposure, the premium due can be accurately calculated. Businesses that may benefit from MPR include construction contractors, amusement parks, and ski resorts.

Policy Setup

The Carrier_Billing_Policy_Account_Common_1_0_0_0 Manuscript defines the interface between policy and billing. This Manuscript contains earned annual premium (EAP), net rate, and exposure information. This information is

mapped from the policy data during the booking process. The data is sent using the <BillingObj.addPolicyTermRq> request and stored in the DC_BIL_PolicyTerm database table for use during subsequent Duck Creek Billing™ processes.

Policy Term Configuration

The following table lists the additional policy plan parameters required to indicate a policy is configured for MPR.

Field Name	Definition
CollectionMSID	The Manuscript ID of the LOB-specific collection screen.
FormMSID	The Manuscript ID of the LOB-specific forms Manuscript.
MPRDueDay	The day the MPR completion is due.
MPRPrdOffset	The number of days before the MPR due date that the MPR form is produced.
PCNPFollowUp	The number of days after the MPR due date that a PCNP follow-up activity is scheduled.
PCNRTrigDays	The number of days after the MPR due date that a PCNR follow-up activity is triggered.
Allow PCNNP	Indicates whether a policy should PCN for non-payment of an MPR receivable.
AllowPCNNR	Indicates whether a policy should PCN for non-reporting.
LastAvailMPRDtDisp	Controls when the last MPR form can be produced for a policy.

Creating an MPR Policy

Because customization is required to utilize the MPR feature, the steps required to create a policy will vary. The following procedure is based on out-of-the-box functionality, and assumes the appropriate Manuscripts have been customized.

To create an MPR policy:

1. Click the **Policy Management** tab.
2. Create an *auditable* policy with a *Monthly* audit period frequency.
3. Enter the policy information.
4. Make note of the policy number before leaving the Policy Management system.
5. On the Submission page, select the **MPR Policy** check box.
6. Click **Complete Issuance**.
7. Click the **Billing** tab.
8. From the Billing Dashboard, in the **I want to find an account by** list, click one of the following options:
 - Account Number
 - Policy Number
9. In the **searching for** text box, enter the policy number noted in Step 4.
10. Click **Search**.

The Account Home page appears and when the MPR form is generated for an account, the activity is included in the **Recent Account Activity** list for the account.

Producing the Form

A forms ManuScript must be created for each line of business. Copy the Billing_MPR_Forms_Line1 ManuScript and adjust the keys and IDs for the target line of business. This ManuScript can be customized with your specific form and data. The EXAMPLE Server® code calls the <billing.printFormsRq> request to initiate the production of the form and creating the activity log entry.

After the first MPR form is created, the next form is automatically scheduled along with a scheduled activity to evaluate PCNR for non-reporting or PCNP for non-payment of a report. EXAMPLE Server® schedules the next MPR activity by using the due day of the account, subtracting the MPRPrdOffset value, and backing up the date if necessary for non-processing days.



To prevent scheduling conflicts, verify the calculated production date is before the last allowed date (LastAvailMPRDtDisp value).

Processing the Returned MPR Form

MPR Report data is entered from the Policy Term Summary page for the appropriate account in Duck Creek Billing™. See *Working with Monthly Premium Reporting* in the *Duck Creek Billing™ User Guide*. The collection ManuScript must be created, and can be modeled after the out-of-the-box sample MPR Collection Line 1 ManuScript that works with BaseProduct.

Pending Cancel for Non-Reporting

A PCNR follow-up is similar to a regular PCN for non-payment. Policy term parameters control whether the policy can PCNR for non-reporting or PCNP for non-payment of the MPR receivable.

Final Audit

When a Final Audit transaction is received for an MPR policy by Duck Creek Billing™, a new receivable transaction for the entire premium amount replaces any existing premium receivables for the policy term.



The only transactions valid in Duck Creek Billing™ following a Final Audit are Revised Audit or Void Final Audit.

For details about entering MPR Report information, see *Working with Monthly Premium Reporting* in the *Duck Creek Billing™ User Guide*.

For more information about configuring an MPR policy, contact your Account Representative.

Integrating with Commerical Lines ManuScripts

The following three ManuScripts are needed to integrate Duck Creek Billing™ with Commercial Lines:

- The Carrier Product Base Workflow ManuScript, which is shipped with Duck Creek Billing™, should be changed to inherit from the DCT_BillingProductBase_Workflow_1_0_0_0.
- The Carrier Product Base Data 7.0.0.0 ManuScript should be changed to inherit from the DuckCreekTech_ProductBaseBilling_Data_7_0_0_0 ManuScript.

- The Carrier Product Base Pages 7.0.0.0 Manuscript should be changed to inherit from the DuckCreekTech_ProductBaseBilling_Pages_7_0_0_0 Manuscript.

For more informatin about using Commercial Lines, contact your account representative.

Configuration Settings

This section contains descriptions, defaults, and valid values of all EXAMPLE Server® user-configurable settings. The Edit Configuration Utility in the Duck Creek root directory (by default C:\Program Files\Duck Creek Technologies\Util>EditConfig.exe) may be used to manually alter these settings.



Improperly adjusting EXAMPLE Server® configuration settings can lead to unintended behaviors or can even prevent EXAMPLE Server® products from functioning. Use caution when changing these settings.

Billing	
Key	Default Value
<i>PoliciesQueue</i>	.\PRIVATE\$\EXAMPLEBILLINGCOMPONENTS0
<i>QueueProcessingUserId</i>	admin
<i>InvoicePath</i>	C:\Program Files\Duck Creek Technologies\Invoices
<i>AgencyStatementsFolderRootPath</i>	C:\Program Files\Duck Creek Technologies\Statements
<i>AgencyStatementsFromEmailAddress</i>	from@domain.com
<i>AgencyStatementsPassword</i>	password55
<i>DirectBillCommissionStatementsFolderRootPath</i>	C:\Program Files\Duck Creek Technologies\DirectBillCommissionStatements
<i>TempWorkingPath</i>	C:\Program Files\Duck Creek Technologies\Temp
<i>ProcessElectronicPayments</i>	True
<i>Use3rdPartyPayments</i>	False
<i>PaymentSiteURL</i>	Null
<i>PaymentSiteUserId</i>	Null
<i>PaymentSitePassword</i>	Null
<i>PaymentSiteReturnURL</i>	Null
<i>EFTTransmissionThrottleBuilder</i>	1
<i>OrphannedCommissionAgentReference</i>	###_DCT_ORPHAN_AGENT_###

<i>DebitChangeNoticePath</i>	C:\Program Files\Duck Creek Technologies\DebitChangeNotices
<i>ExternalDocumentsFolderRootPath</i>	C:\Program Files\Duck Creek Technologies\ExternalDocuments
<i>UseExternalDocumentInterface</i>	False
<i>UseExternalDocumentInBatchMode</i>	False
<i>ScheduledActivityDelay</i>	600000

AgencyStatementsFolderRootPath

Default location for Agency Billing statements.

Default	Valid
C:\Program Files\Duck Creek Technologies\Statements	Any valid location.

AgencyStatementsFromEmailAddress

This setting is the email address used to send Agency Billing statements.

Default	Valid
from@domain.com	Any valid email address.

AgencyStatementsPassword

This setting is the default password that allows a user to unlock the Excel spreadsheet containing an Agency Billing statement.

Default	Valid
password55	Any valid Excel password

DebitChangeNoticePath

Specifies the path for debit change notices.

Default	Valid
C:\Program Files\Duck Creek Technologies\DebitChangeNotices	Any valid directory path.

DirectBillCommissionStatementsFolderRootPath

This setting specifies the location of the commission statements in Duck Creek Billing™.

Default	Valid
C:\Program Files\Duck Creek Technologies\DirectBillCommissionStatements	Any valid file location.

EFTTransmissionThrottleBuilder

This setting specifies use of the EFTTransmissionThrottleBuilder.

Default	Valid
1	• 1 — True • 2 — False

ExternalDocumentsFolderRootPath

Specifies the location of external documents.????

Default	Valid
C:\Program Files\Duck Creek Technologies\ExternalDocuments	Any valid folder location.

InvoicePath

Default path for the location of invoices created by Duck Creek Billing™.

Default	Valid
C:\Program Files\Duck Creek Technologies\Invoice	Any valid location.

OrphanedCommissionAgentReference

Specifies the value for handling agent commission statements for agents no longer associated with an agency.????

Default	Valid
###_DCT_ORPHAN_AGENT_###	??

PaymentSiteURL

This setting specifies the URL for the 3rd party payment processor.

Default	Valid
Null	Any valid URL.

PaymentSiteUserId

This setting specifies the user id for the 3rd party payment processor.

Default	Valid
Null	Any valid user id.

PaymentSitePassword

This setting specifies the password for the 3rd party payment processor.

Default	Valid
Null	Any valid password.

PaymentSiteReturnURL

This setting specifies the return URL for the 3rd party payment processor.

Default	Valid
Null	Any valid URL.

PoliciesQueue

This setting specifies the name of the Message Queue where integration messages between policy and Duck Creek Billing are placed and processed.

Default	Valid
.\PRIVATE\$\BillingIntegration	Any valid message queue.

ProcessElectronicPayment

This setting determines if electronic payment processing is allowed in Agency Billing.

Default	Valid
False	• False — Not allowed. • True — Allowed.

QueueProcessingUserId

This setting is reserved for future use.

ScheduledActivityDelay

Specifies the delay for scheduled activities.

Default	Valid
600000 (milliseconds)	Any valid time entry.

TempWorkingPath

Specifies the location of the temporary working path.

Default	Valid
C:\Program Files\Duck Creek Technologies\Temp	Any valid location.

Use3rdPartyPayments

This setting specifies whether to allow 3rd party payments in Duck Creek Billing™.

Default	Valid
False	• False — Do not allow. • True — Allow.

UseExternalDocumentInterface

This setting specifies whether to use an external document interface.

Default	Valid
False	• True — Use and external document interface. • False — Do not use an external document interface.

UseExternalDocumentInterfaceInBatchMode

This setting specifies whether to use an external document interface for batch processing.

Default	Valid
False	• True — Use and external document interface for batch processing. • False — Do not use an external document interface for batch processing.

Working with Invoices and Installments

Customizing an Invoice

The Manuscript to use for billing forms must be specified in the Invoice Section of the Billing Account Options. For details on how to specify the Manuscript for use, see *Selecting Billing Account Options* in the *Duck Creek Billing™ User Guide*.

See the *EXAMPLE Forms™ Reference Guide* for detailed information about customizing forms in general.

Specifying Non-Processing Days

Non-processing, or blackout days, are specified in the ProcessingBlackoutDates database table. This information is used when determining Target Due Days and Target Draw Days for invoices.

Manually Redistributing Installments

Manual installment redistribution allows customers to have additional time to pay a large amount due, such as final audit premiums. When configured, this feature is accessible from the Installment Schedule page as a link in the Balance column. The link is only available when the Installment Schedule view is expanded, an open balance exists, and the *Account Type Code* does not equal *Agent*.

Installments can be manually redistributed outside the current policy term dates, and any items that have been redistributed outside the policy term are catch-up items on the next installment.



To have the required permissions to manually redistribute an installment, the logged in user must be assigned the role of BillingAgent with the BIL_ManualInstallmentRedistribution privilege. This is an out-of-the-box privilege assigned to the BillingAgent role. For more information on billing roles and privileges, see *Creating a Billing Role* in the *Duck Creek Billing™ User Guide*.

Creating a New EXAMPLE Server® Request

The <BillSchedule.redistributeInstallmentBalanceRq> request must be created to control the redistribution. This request must specify the *BillItemScheduleID*, the earliest installment date for the redistribution, and the number of new installments to create.

Redistributed installments will have a different installment type code than regular installments so they can be easily identified in the database tables. This change also allows credit allocation to work correctly since the usual method of matching debits based on effective and expiration dates will not work with redistributed installments.

The existing credit allocation processing will be changed to search for debits that could be outside the range of installment dates based on their effective and expiration dates.

Working with a Late Entered Policy

Through ManuScript configuration, the *Late Entered Strategy* feature allows a Duck Creek Billing™ user to insert an out-of-cycle draw for an EFT invoice account for a late entered new business policy. Out of the box, without the *Late Entered Strategy* feature enabled, when the scheduled invoice day has already been passed in the current month, a late entered new business policy installment is added to the next scheduled installment for the account, resulting in as much as 31 days of coverage without premium payment.

The *Late Entered Strategy* feature has a positive effect on cash flow and commission authorization, in addition to protecting policy equity by minimizing the amount of time coverage is provided without payment.



The *Late Entered Strategy* is valid for a new business premium by EFT invoice only. Other transactions, such as endorsements and renewals, are scheduled based on the *Late Entered Strategy Code* value on the policy plan associated with the policy. For more information, see *Selecting Policy Plan Options*.

The following table lists the settings that determine the behavior of a late entered new business policy on the installment schedule.

Item	Description	Valid Entries
EFT Draw Offset Days	The number of days between the EFT invoice date and the due date. This value is specified in the Account Settings.	Any positive number between 1 and 31.
Target Draw Date	The day of the month that the EFT draft is drawn from the specified account. This value is specified in the Account Settings.	Any positive number between 1 and 31.
EFT Transmission Offset	The number of days between the EFT Draw Date and the actual draw. This value is specified in the DC_BIL_BankDetails.	Any positive number between 1 and 31.
Policy Effective Date	The date the policy is effective. Specified in the policy management system.	Any valid date.
Policy Issue/ Policy Entry Date	The date the policy is issued or entered into the policy management system. This value is specified in the policy management system.	Any valid date.
EFT Out of Cycle Draw Due to Late Entry	Allows the user to insert an out-of-cycle draw for an EFT invoice account for late entered new business policies to collect payment before the next scheduled EFT draw. This value is specified in the Billing_AccountPlanSetup_Base_1_0_0_0 Manuscript.	True or False.
EFT Out of Cycle Next Draw Offset Days	The number of days before the next regularly scheduled draw date that an out-of-cycle draw can be made. This prevents an out-of-cycle draw from being too close to the regularly scheduled draw. If the value of the parameter is not met, the out-of-cycle draw is not scheduled. This value is specified in the Billing_AccountPlanSetup_Base_1_0_0_0 Manuscript.	Any positive number between 0 and 31.

The following scenario demonstrates the results of installment scheduling that uses the out-of-the-box settings with no late entered policy.

Item	Value	Explanation
EFT Draw Offset Days	10	This example illustrates how scheduling works out of the box, without the <i>Late Entered Strategy</i> feature enabled, and without a late entered policy.
Target Draw Days	16	
Policy Effective Date	January 16	
Draw Transmission Offset Days	2	
Policy Issue/Entry Date	January 6	

The following scenario demonstrates the results of installment scheduling that uses the out-of-the-box settings with a late entered policy.

Item	Value	Explanation
EFT Draw Offset Days	10	In this example, the policy is late entered because sufficient time (10 days) does not exist for the EFT draw between the Policy Issue/Entry Date, the Policy Effective Date, and the Target Draw Date. The installment will bill with the next scheduled installment on February 16, leaving no payment for coverage for between January 16 and February 16. Depending on the value for the <i>Late Entered Strategy Code</i> in the <i>Policy Plan Setup</i> , the missed installment will either be caught-up with the next installment or spread over subsequent installments.
Target Draw Days	16	
Policy Effective Date	January 16	
Draw Transmission Offset Days	2	
Policy Issue/Entry Date	January 7	

The following scenario demonstrates the results of installment scheduling that uses the Late Entered Strategy feature with a value of **0** specified for the *EFT Out of Cycle Next Draw Offset Days* parameter.

Item	Value	Explanation
------	-------	-------------

EFT Draw Offset Days	10	The policy is late entered because sufficient time (10 days) does not exist for the EFT draw between the Policy Issue/Entry Date and the Policy Effective Date. An out-of-cycle installment is needed and is inserted with an invoice date of January 7 and a draw date of January 17. This reduces the time of coverage without premium to two days.
Target Draw Days	16	
Policy Effective Date	January 16	
Draw Transmission Offset Days	2	
Policy Issue/Entry Date	January 7	
EFT Out of Cycle Draw Due to Late Entry	Selected	
EFT Out of Cycle Next Draw Offset Days	0	

The following scenario demonstrates the results of installment scheduling that uses the Late Entered Strategy feature with a value of **7** specified for the *EFT Out of Cycle Next Draw Offset Days* parameter.

Item	Value	Explanation
------	-------	-------------

EFT Draw Offset Days	10	The policy is late entered because sufficient time (10 days) does not exist for the EFT draw between the Policy Issue/Entry Date and the Policy Effective Date. Because the value of the EFT Out of Cycle Next Draw Offset Days is 7 , there is sufficient time for the EFT draw before the next scheduled draw on February 16, so an out of cycle draw is inserted with an EFT draw date of February 9.
Target Draw Days	16	
Policy Effective Date	January 16	
Draw Transmission Offset Days	2	
Policy Issue/Entry Date	January 30	
EFT Out of Cycle Draw Due to Late Entry	Selected	
EFT Out of Cycle Next Draw Offset Days	7	

The following scenario demonstrates the results of installment scheduling that uses the Late Entered Strategy feature with a value of **7** specified for the *EFT Out of Cycle Next Draw Offset Days* parameter and a later Policy Issue/Entry Date than the previous table.

Item	Value	Result
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EFT Draw Offset Days	10	The policy is late entered because sufficient time (10 days) does not exist for the EFT draw between the Policy Issue/Entry Date and the Policy Effective Date. Because the value of the EFT Out of Cycle Next Draw Offset Days is 7, there is not sufficient time for the EFT draw before the next scheduled draw on February 16, so no out of cycle installment is inserted. Depending on the value for the <i>Late Entered Strategy Code</i> in the <i>Policy Plan Setup</i> , the missed installment will either be caught-up with the next installment or spread over subsequent installments.
Target Draw Days	16	
Policy Effective Date	January 16	
Draw Transmission Offset Days	2	
Policy Issue/Entry Date	January 31	
EFT Out of Cycle Draw Due to Late Entry	Selected	
EFT Out of Cycle Next Draw Offset Days	7	

Late Entered Strategy can be used for both single and multiple policy accounts. Missed installments are included in the Installment Schedule, which is accessible from the left navigation panel from the Account Home page.



Change the date filter on the Installment Schedule page to view missed installments. Depending on the value for the *Late Entered Strategy Code* in the *Policy Plan Setup*, the missed installment will either be caught-up with the next installment or spread over subsequent installments.

To use the Late Entered Strategy feature:

1. In the Billing Account Plan Setup Base 1.0.0.0 Manuscript, specify the following values:



If the value in the *EFT Out of Cycle Next Draw Offset Days* is not met, the out-of-cycle draw is not processed. Depending on the value for the *Late Entered Strategy Code* in the *Policy Plan Setup*, the missed installment will either be caught-up with the next installment or spread over subsequent installments.

- Set the **EFT Out of Cycle Draw Due to Late Entry** parameter to **True** (1).
- Set the **EFT Out of Cycle Next Draw Offset Days** parameter to specify the number of days before the next scheduled draw that the out-of-cycle draw should be processed. This setting can be **0** which does not specify a number of days and means the installment is automatically processed.

2. **Save** the Manuscript changes.

Working with Commissions

Commission Plans

Before premium is issued, an overall commission rate or amount must be specified. The commission can be a percentage or a flat amount, and it can vary by policy term or be the same across all policy terms. The commission plan can vary by the following policy attributes:

- Agent Reference
- Line Of Business
- Master Company
- Issue System
- Product Code
- Inception Date
- Policy Term Effective Date
- State

Five customer-definable keys are also used to determine the correct commission plan. One of the keys can be used to indicate the selected coverage and premium class combination. The same definable key can be used for a separate line of business or product in more than one commission plan.

When a policy is rated, the commission rate or flat amount is determined by performing a look-up based on policy attributes specified in the creation of the commission plan, including the customer definable keys. The search begins at the lowest level of the hierarchy to obtain the correct commission plan, as well as the overall rate of commission or flat amount.

Commission plan changes are supported through the use of effective and expiration dates, which allows new policies to use new commission rates or flat amounts and commission distributions while existing policies use prior commission rates or flat amounts and distributions.

The commission distribution specifies the entity type that will receive the commission distribution and the amount of the distribution. Using the entity tiers created in EXAMPLE UserAdmin™, commission plans can be inherited from a higher agent tier to a lower tier. A commission plan that is inherited from a higher tier can only be changed at the tier where the commission plan was created.

Commission plans associated with an account are listed on two commission plans pages:

- **Active Commission Plans** – Enables the user to add a plan, maintain an existing plan, and view all active commission plans associated with an agent's account.
- **Expired Commission Plans** – Displays the details of expired commission plans that were previously associated with an agent's account.

To view the commission plans associated with an account:

1. In the left navigation panel of one of the following pages, click **Commission Plans** :
 - Account Home

- Installment Schedule
- Make A Payment
- Items in Suspense
- Hold Billing Activities

2. To view the active commission plans, click **Active Commission Plans**.
All of the Inherited Plans and Agent Specific Plans associated with the selected account appear.



Inherited plans are view-only. Any changes to a commission plan must be made at the level where the plan is created.

3. To view the details of an inherited plan, click **Details**.

To add a new commission plan:

1. From the Active Commission Plans page, click **Add New Commission Plan**.
The Commission Plan Details page appears.
2. Accept the default selections, or use the dropdown list or text box to enter any different values.
3. Enter the plan name in the **Description** text box.
4. In the **Terms** section, below the *Commission Rate* heading, click one of the following options in the **Type** list:
 - Percentage
 - Flat Amount
5. In the **Value** text box, enter the appropriate amount.
6. Below the **Commission Distribution** heading, click one of the available options in the **Entity Type** list.



The available options are determined by ManuScript customization.

7. In the **Percentage** or **Amount** text box, enter the appropriate amount.



Enter percentages as numbers carried out to two decimal points. For example, 10 percent would be entered as 10.00.

8. After entering the value, click the page outside of the text box to refresh the page, or tab outside the text box.
9. (Optional) To add an additional distribution for the commission amount, click **Add New Distribution**.
10. (Optional) Repeat Steps 6 and 7 for each additional distribution.
11. (Optional) Click **Delete Distribution** to remove a distribution from the commission plan.
12. Click the page outside of the text box to refresh the page.



The percentage or amount for other entity types will need to be changed to match the specified commission rate amount.

13. Click **Save New Plan**.



The **Save New Plan** option will not appear until the values entered in the **Commission Percentage** or **Amount** text box balances with the sum of the distribution percentage or amount.

The Active Commission Plans page appears.



The Save All Changes option is not enabled unless the plan has been changed.

14. Click **Save All Changes** to save the plan changes.
15. (Optional) Click **Edit Plan** to edit the new plan.
16. (Optional) Click **Delete Plan** to delete the new plan.

To add a renewal term:

The Terms section of the page refreshes, and displays separate Commission Rate and Commission Distribution options for new business and renewal terms.

1. Click **Add Renewal Term**.
2. In the new business section, below the **Commission Rate** heading, click one of the following options in the **Type** list:
 - Percentage
 - Flat Amount
3. In the **Value** text box, enter the appropriate amount.
4. Below the **Commission Distribution** heading, click one of the available options in the **Entity Type** list.
5. In the **Percentage** or **Amount** text box, enter the appropriate amount.
6. After entering the value, click the page outside of the text box to refresh the page, or tab outside the text box.
7. (Optional) Click **Add New Distribution** to add another entity type.
8. In the **Percentage** or **Amount** text box, enter the appropriate amount.
9. After entering the value, click the page outside of the text box to refresh the page.
10. Click **Save New Plan**.
11. Click **Save All Changes** to save the plan changes.
12. (Optional) Click **Edit Plan** to edit the new plan.
13. (Optional) Click **Delete Plan** to delete the new plan.

To edit an existing agent specific plan:

1. Click **Edit Plan**.
The Commission Plan Details page appears.
2. Edit the desired details.
3. Click **Save Changes**.
4. (Optional) Click **Discard Changes** to restore the previously saved plan.
5. Click **Return To Plan List** to return to the Active Commission Plans page.
6. Click **Save All Changes** to save the changes made on the previous page.

7. Click **Undo All Changes** to discard the changes made on the previous page.

To view the expired commission plans associated with an account:

1. Click **Expired Commission Plans**.
 2. (Optional) Click **Details** to view commission plan details.
 3. Click **Return To Plan List** to return to the Commission Plan Details page.
- Expired commission plans are retained on an account, and an audit trail is automatically maintained.

Configuring Commission Options

Commission rate and eligibility are determined in the policy system and transmitted to Duck Creek Billing™ on the policy term premium record. Records are written to database tables based on the commission scheme specified in the Policy Plan Setup option. The CommissionStatementSchedule database table maps an agent to the statement schedule and produces statements based on the data in the table.

The <BillCommission.buildCommissionStatementRq> request populates the following two tables:

- **DC_BIL_DirectBillCommissionStatementSummary** — Contains information needed by a check-writing interface.
- **DC_BIL_DirectBillCommissionStatementDetails** — Contains details for statement make-up to generate printable statement.

Most data in these tables is stored in an XML column.



Commission statement requests should be initiated serially from a single source and processed by a single server to avoid possible race conditions between multiple threads and/or servers when writing to the disbursements file.

Commission processing uses two requests:

- The <BillCommission.buildCommissionStatementRq> request uses the agencyId, statementDate, and printDirectory attributes.
- The <BillCommission.printCommissionStatementRq> request uses the statementId and printDirectory parameters to generate printable statements. This request initiates a report using SQL Server 2005 Reporting Services to produce statements.

Reports are output as PDF files to the directory specified in the printDirectory attribute. If the print directory is not passed in as an attribute, the DirectBillCommissionStatementsFolderRootPath setting in the Server.Config file is used.

The statement file name will be DirectBillCommissionStatement_Agency_[Agency]_[StatementDate]_[StatementId].pdf.

Customers are responsible for printing and sending the statements, or maintaining the files.

Database Components

The following table includes a list of the database tables used to configure commissions.

Table Name	Definition
------------	------------

DC_BIL_CommissionAuthorization	The AgencyRollupReference column contains the general agency for the policy at the time of the commission authorization. The AuthorizationDateTime column contains the exact date and time that the commission is authorized.
DC_BIL_DirectBillCommissionStatementSummary	This database table stores the summary information for each commission statement. This table provides the information needed by the interface to the customer's external check writing system. To support extensibility and interface flexibility, most of the data is contained within a single XML data column.
DC_BIL_DirectBillCommissionStatementDetail	This database table holds the detailed information for creating the statement. The information in this table is used to generate a commission statement for a given agency. Most of the data is contained within a single XML data column.

Configuring Commissions

The commission scheme for a policy is specified from the Policy Term Settings for the appropriate account. See *Working with Commissions* in the *Duck Creek Billing™ User Guide*.

In addition to selecting the commission scheme option, ManuScript customization is required for commission configuration.

To customize the appropriate ManuScript:

- Specify the values for the following two fields in the appropriate ManuScript:
 - CommissionAmount
 - CommissionPercent

The following table describes the valid values for the two ManuScript fields required for commission generation.

Field Name	Definition	To Disable Commissions
CommissionAmount	Calculation that determines commission amount by multiplying the value in the PolicyOutput.PremiumChange field by the value in the PolicyOutput.CommissionPercent field.	Set the field type to Constant with a value of 0 .
CommissionPercent	The value multiplied by the PolicyOutput.PremiumChange to determine commission amount.	Set the value to zero (0). Default is .10 (10 percent).

- Save the ManuScript changes.

Available Commissions Plans

The following table lists the out-of-the-box commission payment strategies available in Duck Creek Billing™.

Table Name	Definition	Setting
Paid on Written	Payment is not required before commission is released. The entire commission amount is authorized when the receivable is posted. The commission authorization date is the activity date of the policy transaction. Debit receivables create an authorization record. Credit receivables generate a de-authorization record.	POW1
Paid on Effective	The commission is authorized for each policy transaction at issuance. Payment is not required before the commission is released. The commission authorization date is the latter of the effective date or the activity date of the policy transaction.	POE1
Paid on Paid Pro Rata	Payment is required before commission is released. The amount of commission that has been paid to date for a given receivable is evaluated. Next, based on the payment activity, the amount of commission that should have been paid is evaluated. The difference is the commission amount that is authorized. The activity date of the payment is used as the commission authorization date for all debit policy transactions. All other policy transactions use the activity date of the policy transaction as the commission authorization date. When premiums are written off, the corresponding commission amount is still authorized.	POP1
Paid on Paid First Payment	The entire commission amount is paid when the first payment is applied to the receivable. Receiving one valid payment on the policy term releases the commission for that policy term. The activity date of the payment is used as the commission authorization date for a new business or a renewal policy transaction. All other policy transactions use the activity date of the policy transaction as the commission authorization date.	POP2
Paid on First Paid	The commission is released following the receipt of one valid payment on the policy term. The latter of the policy transaction effective date or the activity date of the payment is used as the commission authorization date for a new business or renewal policy. All other policy transactions use the latter of the policy transaction effective date or the activity date of the policy transaction as the commission authorization date.	POP4

Working with Agent Commissions

Agent Commission accounts are used to specify how commissions are paid to the various levels that are determined when the commission account is created. The commissions features in Duck Creek Billing™ can be customized through ManuScript configuration to accommodate a variety of commission plans and payment schemes.

The first step for working with agent commission is creating the multi-level commission feature using the role-based security features in EXAMPLE UserAdmin™. Next, a Direct Bill Commission account is created to contain the details for how and when the commission is collected and distributed to the agent. Part of the account creation includes specifying the format that will be used to generate commission statements for the account. After the hierarchy is created, the agent's account is setup, and statement format is specified, the final step for working with agent commission is actually creating commission plans to associate with the agent's account. Commission plans can be created, edited, or deleted using the Commission Statement Maintenance page. Commission plans can be inherited through the hierarchy created in EXAMPLE UserAdmin™. The following topics explain each of these required steps in detail.

Working with Multi-level Commisisions

The multi-level commission feature is enabled by using the role-based security features in EXAMPLE UserAdmin™.



Recommended Reading:

The *EXAMPLE UserAdmin™ Overview* describes the basic concepts of role-based security and how the EXAMPLE Platform® uses EXAMPLE UserAdmin™, and the *"Initial Setup"* in the *EXAMPLE UserAdmin™ User Guide*, which describes the process of working with entities, roles, and users.

EXAMPLE UserAdmin™ may be used to set up and define the entities, roles, users and hierachy (parent/child relationships for commission distribution and plan inheritance. Commissions may be distributed at one or multiple levels in the hierarchy as specified in the commission plan that is created for an account. Commission plans can be created at one tier, and then be inherited by a lower tier, but they cannot be created at a lower tier and then be inherited at a higher tier.

A policy's commission plan is determined when the policy is entered. The policy administration system retrieves the commission plan ID and sends it to Duck Creek Billing™ where it is stored in each BillItem record. When the commission is distributed, it pulls the appropriate commission plan using the commission plan ID. The necessary keys are used to search from the lowest tier of the hierarchy to obtain the correct commission plan and distribution. The commission distribution includes each recipient ID, as well as the recipients' distributed commission percentage.

When a search for a commission plan begins, it is based on both the fixed and variable policy attributes specified in the commission plan. The search begins at the lowest level of the hierarchy and moves upward. If no match is found at the first level, the search moves to the next tier in the hierarchy. If a match is found at this tier, the commission plan is returned. The search process continues through all the tiers, and if no match is found at the highest tier, an error is returned.

For a given commission plan, the user may also specify different rates and distributions for new business and renewals.

Overview for Creating a Multi-Level Commission Plan

The following table includes the process for creating a multi-level commission plan.

Step	Description
1	Determine the required hierarchy.
2	Determine whether to use existing entity types or create new entity types for your commission structure.
3	To create new entity types, see <i>Working with Entity Types</i> .
4	Determine what entities need to be added to the entity types.
5	To create new entities, see <i>Working with Entities</i> .
6	Associate the required roles with each entity.
7	If additional roles are needed, see <i>Working with Roles</i> to create new roles.

8	Select the required Lines of Business for each entity. For more information, see <i>Working with Product Authorizations</i> .
9	Select the parent for the entity. For more information, see <i>Working with Entities</i> .
	 This step is not required for the top tier of the hierarchy.

Creating an Agent Commission Account

Users create a Direct Bill Commission account type specifically to collect and distribute agent commission. This account contains information for each specific agent about that agent's preferred statement format and commission payment method.

The following ManuScripts are used in the customization of the available account options:

- Billing_Policy_Account_Common_1_0_0_0.xml
- Billing_Commission_Plans_1_0_0_0
- Carrier_Billing_Commission_Plans_1_0_0_0
- Billing_AccountPlanSetup_TemplateDirectBillCommission_1_0_0_0
- Agent_Plans_1_0_0_0
- Carrier_Agent_Plans_1_0_0_0

Commission payments to agents can be made by check or EFT. When payments are made by check, a file is automatically created for use with the Duck Creek Billing™ disbursement interface. When the **EFT** option is selected, the user must enter bank details for the agent, and a commission payment is processed like any other disbursement. Any amount that is not paid by EFT or check is returned and applied back to the agent's commission statement to be included with the next payment.



The Commission Account privilege is required for access to the Account Home page for a direct bill commission account. For more information, see *Creating a Billing User*.

Creating an Agent Account

To create an agent account:

1. On the Billing Dashboard, click **Create A New Account**.
2. In the **Bill Account Class** list, click **Direct Bill Commission**.
3. Click **Search For An Existing Party**.
4. In the **I'm looking for** list, click **Agency**.
5. In the **I want to search by** list, click **Reference Number**.
6. Enter the reference number for the appropriate EXAMPLE UserAdmin™ entity.
7. Click **Search For Party**.
8. Click the appropriate existing party.
The Create New Account page appears.

9. Click **Create New Account**.
10. In the **Commission Statement Format** list, click the appropriate statement format or formats.
11. In the **Commission Statement Frequency** list, click the appropriate day.
12. In the **Commission Statement Payment/Collection Method** list, click one of the following options:
 - Check
 - EFT
13. (Optional) Enter bank account details for EFT payments.
14. Click **Complete**.
15. (Optional) Click **Commission Plans** in the left navigation panel of the Account Home page to create a commission plan for this account.
16. (Optional) Click **Account Settings** in the left navigation panel of the Account Home page to change account settings.

Working with the Commission Statement Maintenance Page

The Commission Statement Maintenance page provides quick user access for working with commission statements for an account.

The Commission Statement Maintenance page allows the user to:

- View the details of a pending commission statement
- Stop or resume the production of a pending commission statement
- Create an ad hoc (on demand) commission statement
- Manually adjust amounts on the pending commission statement

The commission statement includes the following information:

- Policyholder Name
- Gross Premium Amount
- Commission Rate or Flat Amount
- The Seller (Producer) Reference
- Seller (Producer) Name
- Authorized commission that has not been paid for the current commission statement period
- Authorized commission that has been reversed, or clawed back, either manually or automatically

(An automatic adjustment is triggered in response to a policy premium transaction)

(A manual adjustment is initiated by a user and is not triggered in response to a policy premium transaction)

- Authorized commission that has been paid, but the payment was returned

The Commission Statement Maintenance page includes only items associated with the current statement period. A debit balance carries forward as a single entry into the next commission statement period. To reduce or eliminate a debit balance, an agent payment can be entered from the Make A Payment page for the appropriate account, or be entered as a single payment from the Billing Dashboard.

The Commission Statement Maintenance page is accessible from the left hand navigation panel of the following pages:

- Account Home
- Installment Schedule
- Make A Payment
- Items in Suspense
- Hold Billing Activities

Viewing a Pending Commission Statement

The following information appears in the Summary section of the Commission Statement Maintenance page:

- Agent Reference Number
- Agent Name
- Statement Period
- Previous Statement Balance
- Payments Made/Received
- Current Activity
- Current Statement Balance

Holding or Resuming Statement Production

When the production of a commission statement is placed on hold, it is for an indefinite period of time. The hold must be manually released to resume the commission statement production. Holding and resuming statement production is added to the **Recent Account Activity** list for the agent's account, but not all the details are listed. If an agent is terminated or canceled, commission statement generation does not automatically stop, but can be manually suspended using the hold statement feature.

To place a commission statement on hold:

1. On the Commission Statement Maintenance page, in the **Hold Reason** list, click the appropriate option.



Out of the box, the available options are:

- Reason 1
- Reason 2
- Reason 3

The text for available options may be customized in the
Carrier_Billing_Commission_Statement_Authorization_Maintenance_1_0_0_0.xml Manuscript.

The **Hold Statement** button appears.

2. Click **Hold Statement**.

The statement will remain on hold until statement activity is manually resumed.

To resume statement production:

- Click **Resume Statement**.

Requesting an Ad Hoc Statement

Commission statements are produced on a specified scheduled, usually monthly. Commission statement periods do not have overlapping days. For example, the dates included in the commission begin with January 1, 2011, at 12:00 a.m. and ends with January 31, 2011 at 11:59 p.m.

The statement period for an ad hoc statement is specified by when the statement is generated. Ad hoc statements include all the activity that has occurred since the previous statement was produced. After an ad hoc commission statement is produced, the next commission statement is produced according to the original schedule.

An ad hoc statement does not appear for viewing, but is produced in the format or formats specified in the commission plan setup, and is automatically placed in the appropriate statement folder. An Archive folder is also created and contains a copy of each statement format generated for an account. By default, this folder location is: `C:\Program Files\Duck Creek Technologies\DirectBillCommissionStatements`.

To request an ad hoc statement:

- On the Commission Statement Maintenance page, click **Request Ad Hoc Statement**.
The statement is automatically placed in the archive folder and all other appropriate folders, based on statement format.

Adjusting the Commission Statement

Commission adjustments can be made at the account or policy level. New entries can be added to a pending commission statement, including both negative and positive entries.

- A negative entry reflects a new claw back authorized commission. A claw back is associated with a specific policy.
- A positive entry reflects payables that are not generated from anywhere, but rather are created and result in additional commission for the agent can be added. Examples could include a bonus or repayment of an expense incurred by the agent.

Adjustments can be made only on pending commission statements. A commission statement that has been paid cannot be adjusted. Any necessary adjustment must be made on a subsequent statement.

Adjustments to a pending commission statement are recorded in the General Ledger report which is accessible from the Billing Dashboard. The Commission Activity report is also available from the Billing Dashboard.

To adjust a commission plan:

Commission Adjustment is selected by default in the **Reason** list.

1. On the Commission Statement Maintenance page, enter the appropriate policy number in the **Policy Number** textbox below the **Statement Entries** heading, and then click **Search**.
The account number, policy holder name and policy term information appear.
2. Click **Add Commission Adjustment**.
The Add Commission Adjustment window appears.
3. Enter the appropriate date, or use the calendar button `calendar button.png` to select a date in the **Authorization Date** list.



The adjustment date must be the current date or later.

4. Enter the adjustment amount in the **Authorized Amount** textbox.



For dollar and cents amounts, enter the full amount including the decimal point.
For a debit, enter a minus sign (-) before the amount. Entering a dollar sign (\$) is not required.

5. Click **Add**.

Details of the adjustment appear on the Commission Statement page, and the commission statement summary at the top of the page is updated based on the adjustment amount.

Multi Format Commission Statements

Commission statements are available in the following formats:

- Adobe PDF (PDF)
- Microsoft Excel (XLS)



Compatible versions of Microsoft Excel and SQL Server are required. SQL Server 2005 requires Microsoft Excel 2003.

- Comma Separated Value (CSV)
- XML

To provide flexibility, more than one format may be selected for the same commission statement. For example, by selecting both PDF and XLS formats, the user can print the PDF statement for mailing, and attach the XLS statement to an email without additional work.

When a statement is produced, a folder is automatically created for each selected statement format. An Archive folder that includes a copy of each statement format is also created. By default, the folder location is `C:\Program Files\Duck Creek Technologies\DirectBillCommissionStatements\`. The commission statement folder location is specified during the Duck Creek Billing™ installation process, and is configurable.

To specify the statement format:

1. From the Account Home page, in the left navigation panel, click **Account Settings**.
2. In the **Commission Statement Format** section, click the appropriate options.



The **Adobe PDF** format check box is automatically selected, and cannot be cleared because it is required to enable the ability to view commission statements online.

3. Click **Save Changes**.

Viewing Commission Statements Online

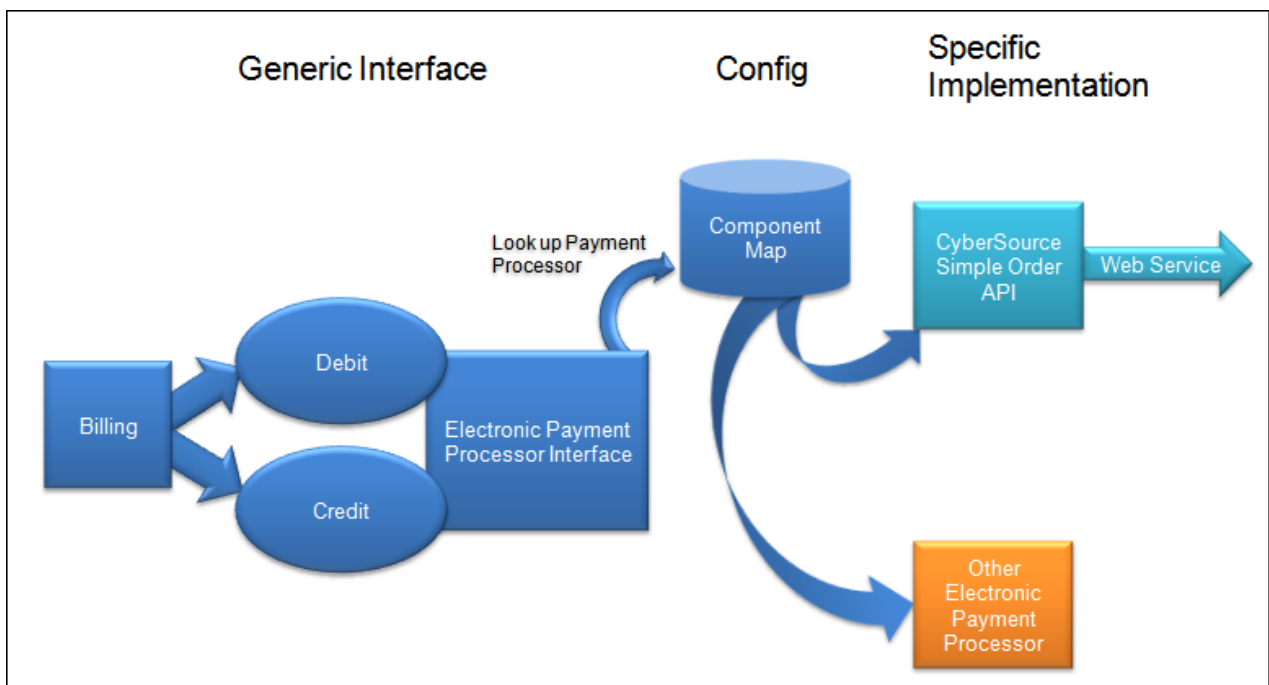
The View Statement link is added to the **Recent Account Activity** list when a commission statement is generated for an account. This includes scheduled monthly statements and ad hoc statements. The link allows the user to view a PDF version of the statement from the Account Home page. Regardless of the statement format option or options selected for the account, the online statement is only available as a PDF.

To view the statement:

- Click the **View Statement** button View Statement Image.png for the appropriate item on the **Recent Account Activity** list.
The File Download dialog box appears, and allows the user to view or save the PDF file.

Working with the Third Party Payment Interface

The integration process between Duck Creek Billing™ and a third party payment interface to process credit card, debit card, and electronic checks is shown below.



Configuring Third Party Payment Processing

Duck Creek Billing™ can be integrated with a third party payment processor.



Lock box functionality must be configured prior to successful deployment of the Third Party Payment Interface. See *Configuring the Lockbox*.

The Third Party Payment Interface is disabled by default.

To enable the Third Party Payment interface:

1. Open the **Edit Configuration** utility.
2. In the **Virtual Directory** list, click the **Express** Virtual Directory.
3. Click the **Web Config** tab.
4. Browse to **Use3rdPartyPayments**.
5. In the **value** field, change the setting to **True**.
6. Browse to the **ExternalPaymentsInterface**.
7. In the **value** field, set the path to the **ExternalPaymentInterface.xml** file.



The default location for this file is: <installed dir>\Express\App_Data\ ExternalPaymentInterface.xml

8. Click **Save**.

The External Payment Interface must also be configured. The ExternalPaymentInterface.xml file contains a node called ExternalPaymentInterface. EXAMPLE Express® uses the data configured in this section to build the URL and parameters to submit to a Third Party Payment Provider.

To configure the External Third Party Payment Interface:

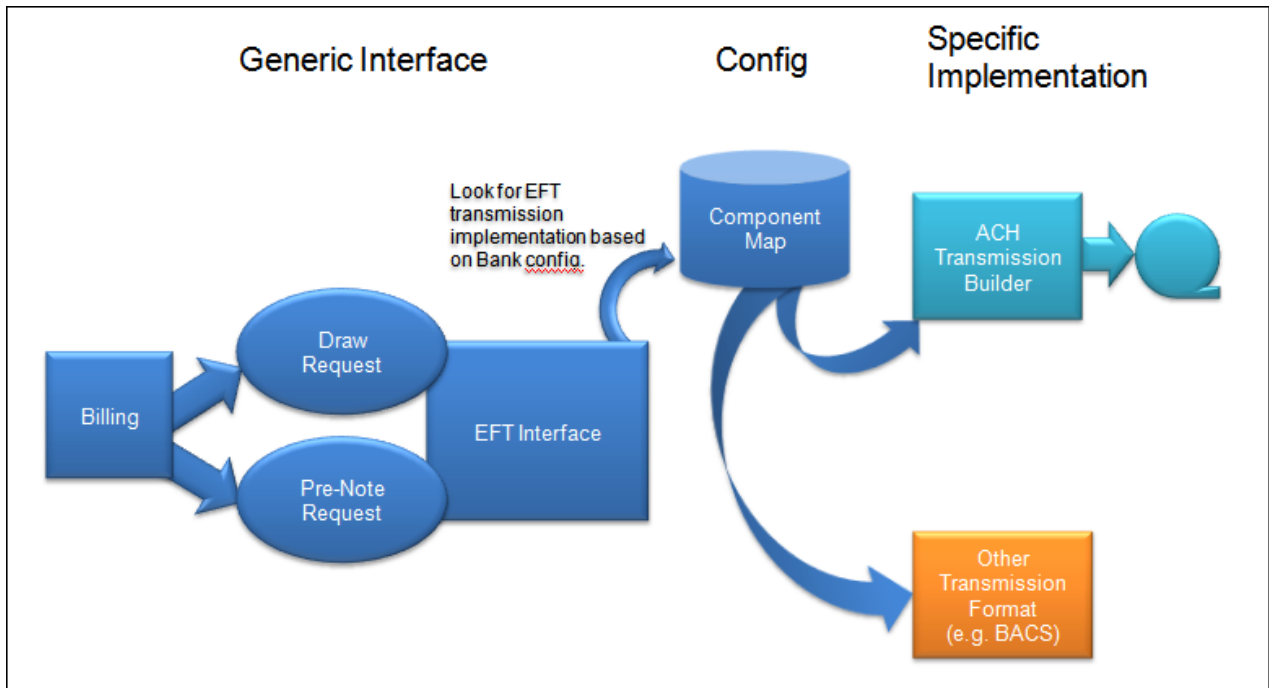
1. Open the **Edit Configuration** utility.
2. In the **Virtual Directory** list, select the **Express** Virtual Directory.
3. Click the **Server Config** tab.
4. Expand **Billing**.
5. Configure the parameters listed in the following table.

Name	Type	Required	Description
PaymentSiteURL	String	Yes	Specifies the URL of the Third Party Payment Provider.
PaymentSiteUserId	String	Yes	Specifies the UserId for the Third Party Payment site.
PaymentSitePassword	String	Yes	Specifies the password for the Third Party Payment site.
PaymentSiteReturnURL	String	Yes	Specifies the return URL after the payment is completed.

Specifying the parameters in the table will direct traffic to and from a third party payment site. For assistance with third party payment processing, contact your Account Manager.

Working with Electronic Funds Transfers (EFT)

The integration between Duck Creek Billing™ and Electronic Funds Transfer (EFT) payments is illustrated below.



EFT works on a negative response basis which means payment is assumed unless the bank returns a rejection notice.

EFT payments are based on the following assumptions:

- The account is not in a stop or hold status
- Policies under the account that are in a pending cancel status are not included in the draw amount being billed for the current billing period.
- The account is in an active EFT status or pending EFT status.

Customization is required to use this feature. For more information about integrating EFT payments and Duck Creek Billing™, contact your Account Manager.

Configuring the EFT Maintenance Privilege

A user must be authorized to change EFT draw dates or cancel a pending EFT draw. The following procedure details the steps required to create and associate this privilege with a Duck Creek Billing™ user. If a user does not have the required permission or the draw amount exceeds the user's limit, the EFT pages are read-only.



For information about using this feature, see *Working with EFT Maintenance* in the *Duck Creek Billing™ User Guide*.

To create the required privilege for authorizing disbursements and associate that privilege with a user:

1. Open EXAMPLE UserAdmin™, and in the left navigation panel, click **Add a role**.
2. Enter a role name and description in the appropriate text boxes.

3. Click **Select General System** and then select the appropriate privileges.
4. Click **Select Policy Administration** and then select the appropriate privileges.
5. Click **Billing Administration** and then select the appropriate privileges.
6. Click **Author Administration**, select the appropriate privileges, and then click **Finish**.
7. Click the **View Entities Associated with this role** button  for the newly created role.
8. Click **Add more entities to role**.
9. Enter an entity name in the **Enter Name** text box and then click **Search**.
10. Select the appropriate entities, and then click **Add roles to these entities**.
11. Click **Manage role/privilege associations** in the left navigation panel.
12. In the **Role** list click the appropriate role, and then in the **Privilege** list, click **BIL_EftMaintenance**.
13. Enter **EFTMaintenance** in the **Name** text box, enter the appropriate value in the **Value** text box, and then click **Insert**.
14. To create a new user or associate this role with an existing user, see [Working with Users](#).

Working with an Automated Notice of Change

The bank account information used by EFT invoice accounts is automatically updated or created in Duck Creek Billing™ in response to messages from a bank or payment processing service.

When Duck Creek Billing™ receives a message to change the bank details for an account, the changes are made automatically. When the changes are successfully processed, a message is added to the **Recent Account Activity** list for the account.



To prevent disclosure of confidential information, details are not included in the message on the **Recent Account Activity** list. For example, a routing number change would not include the actual routing number, but instead would only indicate the routing number has changed.

An error message is created in the Billing Error Queue if bank account changes are not processed by Duck Creek Billing™.

Changes are effective when they are processed and do not affect any previous transactions. If an account is on hold when a change message is received, Duck Creek Billing™ processes the message and the account is updated. Details can be edited after they are automatically updated.

The following table lists the specific ACH file format requirements for the records affected by an Automated Notice of Change.

Record Name	ACH File Format Requirements
-------------	------------------------------

Batch Header	• Entry class code • Debit date • Bank routing transit number • Batch header number  There is only one batch header record for each debit date processed, but there may be multiple batch header records per tape.
Detail Record	• Amount • Trace number for each EFT rejection  There is one detail record for each transaction being rejected.
Detail Addenda Record (Includes Rejection and Change Records)	• Rejection records support rejections of assumed payments and prenotes. • Change records support change transactions.  Each record must include a detail addenda for either rejection or change.
Batch Control Record	• Number of detail records within a batch • Total debit amount • Bank routing transit number • Batch number  There is one batch control record for each batch header record.

The following table lists the value and content of the Batch Header record.

Length	Position	Field Name	Description
1	1	Record type	A required 1-character field which indicates the type of input record. The valid type is 5.
3	2 - 4	Service class code	A 3-position numeric field which identifies the general classification of dollar entries to be exchanged. For example, debits, credits, or mixed debits and credits.
16	5 - 20	Company name	A 16-position character field established by the originator for further identifying the source of the entry, and for descriptive purposes for the receiver.

20	21 - 40	Company discretionary data	A 20-position character field which allows originators to include codes of significance only to them, and to enable specialized handling of all subsequent entries in that batch.
10	41 - 50	Company identification	A 10-position character field which identifies an originator.
3	51 - 53	Standard entry class code	A required 3-position character field which permits the following entries to be distinguished: • RET for rejections • COR for changes • CCD for commercial deposit files • PPD for personal deposit account files
10	54 - 63	Company entry description	A 10-position character field used to describe the purpose of the entry to be displayed back to the receiver.
6	64 - 69	Company descriptive date	A 6-position character field used by the originator as the date it would like to see displayed to the receiver for descriptive purposes.
6	70 - 75	Effective entry date	A required 6-position character field which is the date specified by the originator on which it intends a batch of entries to be debited.  The format is MMDDYY for COR and RET.
3	76 - 78	Settlement date	A 3-position numeric field which identifies the scheduled settlement date for a batch of entries.
1	79	Originator status code	A 1-position character field which refers to the originating depository (financial institution) initiating the entry.
8	80 - 87	Originating DFI identification	A required eight 8-position character field which identifies the DFI originating entries (bank routing transit number).
7	88 - 94	Batch number	A required seven 7-position numeric field assigned in ascending sequence for each batch by the sending point in a given file of entries.

The following table lists the value and content of the Detail record.

Length	Position	Field Name	Description
1	1	Record type	A required 1-character field which indicates the type of input record. The valid type is 6.

2	2 - 3	Transaction code	A required 2-position character field which identifies various types of debit and credit entries. Valid codes are: • 21 – DDA Credit • 26 – DDA Debit • 31 – Savings Credit • 36 – Savings Debit • 27 – DDA Prenote • 37 – Saving Prenote
8	4 - 11	Receiving DFI identification	An 8-position character field which is the standard routing number.
1	12	Check digit	A 1-position character field which is a sub- field within individual DFI identification.
17	13 - 29	DFI account number	A 17-position character field which identifies the receiving depository financial institution's customer identification.
10	30 - 39	Amount	A required 10-position numeric field which contains the amount associated with the detail record.
15	40 - 54	Individual identification number	A 15-position character field which contains the accounting number by which the receiver is known to the number originator.
22	55 - 76	Individual name	A 22-position character field which provides additional identification for the receiver.
2	77 - 78	Discretionary data	A 2-position character field used by the originating depository financial institution to include codes used for specialized handling of the entry.
1	79	Addenda record indicator	A 1-position numeric field used to indicate the existence of the addenda record.
15	80 - 94	Trace number	A required 15-position numeric field which uniquely identifies each entry within a batch.

The following table lists the value and content of the Detail Addenda Record for Rejection.

Length	Position	Field Name	Description
1	1	Record type code	A required 1-position character field which indicates the type of input record. The valid type is 7.

2	2 - 3	Addenda type code	A 2-position character field which defines the specific interpretation and format for the addenda information contained in the same record. The valid type for Rejections is 99.
3	4 - 6	Return reason code	A required 3-position character field which is a standard code used to describe the reason for returning an entry.
15	7 - 21	Original entry trace number	A required 15-position character field which is the trace number that was originally included on the entry being rejected.
6	22 - 27	Date of death	A 6-position character field which is the date of death for those entries being returned for reason of death.
8	28 - 35	Original receiving DFI ID	An 8-position character field which is the receiving DFI Identification as originally included on the entry.
44	36 - 79	Addenda information	A 44-position character field used to relay explanatory information required with the use of certain return reason codes.
15	80 - 94	Trace Number	A 15-position numeric field which uniquely identifies each entry within a batch.

The following table lists the value and content of the Detail Addenda Record for Change.

Length	Position	Field Name	Description
1	1	Record type code	A required 1-position character field which indicates the type of input record. The valid type is 7.
2	2 - 3	Addenda type code	A 2-position character field which defines the specific interpretation and format for the addenda information contained in the same record. The code for Notices of Change is 98.
3	4 - 6	Change code	A required 3-position character field which is used to relay corrected customer information back to the originator of that entry.
15	7 - 21	Original entry trace number	A 15-position character field which is the trace number that was originally included on the entry being rejected.
6	22 - 27	Reserved	A 6-position character field which is reserved for use by the ODFI.
8	28 - 35	Original receiving DFI ID	An 8-position character field which is the receiving DFI identification as originally included on the entry.

29	36 - 64	Corrected data	A required 29-position character field which is used to relay corrected customer information back to the originator of that entry.
15	65 - 79	Reserved	A 15-position character field which is reserved for use by the ODFI.
15	80 - 94	Trace Number	A 15-position numeric field which uniquely identifies each entry within a batch.

Working with a Bank Details Modulus Check

Through customization and integration with Duck Creek Billing™ Automated Clearing House (ACH) and Banker's Automated Clearing Services (BACS) verification can be utilized to reduce the number of rejected transactions.

Users can design a custom workflow using the Modulus Check Request and Account Setup provided in Duck Creek Billing™

For more information about specific customization and integration, contact your Duck Creek Account Manager.

Working with Refunding to a Payment Source

The Refund to Payment Source feature allows refunds to be made automatically to the source of payment. It requires configuration and integration with custom interfaces.

To utilize the Refund to Source feature:

1. From the Account Home page for the appropriate account, click **Account Settings** in the left navigation panel.
2. In the **Disbursement** section, select the **Refund To Source** check box.
3. Clear the **Hold Disbursement Indicator** check box to allow automatic disbursements.

Although no report of rejected refunds is available, a list or report can be created by the return file or a transmission from the bank. Regardless of the reason, all rejected refunds are placed in the **Items In Suspense** list. The refund must be manually re-suspended in the associated bill account, and can either be resubmitted to the source for refunding or be refunded manually.

This feature supports both the ACH EFT format which is utilized in the United States, and the BACS EFT format which is utilized in the United Kingdom.

The following table lists the custom interfaces required for refund processing.

Method	Interface
Check	Disbursement Interface
Credit Card or Debit Card	Electronic Payment Interface
EFT	EFT Transmission Builder



Only a user with the Override Refund To Source privilege can specify that a refund be made in a format other than the payment source or group multiple payments into a single refund.

If the user has the required permissions, the Refund to Check and Refund to EFT check boxes are enabled.

To override the source of payment for a refund:

The Items in Suspense page appears.



To perform the same action on multiple items in suspense, select all the appropriate check boxes.

The disburse funds page appears and displays each select suspense item, the amount, and the refund method.

The Refund To Source page appears with the selected payee displayed.

The party data system search appears.

The Refund To Source page appears.



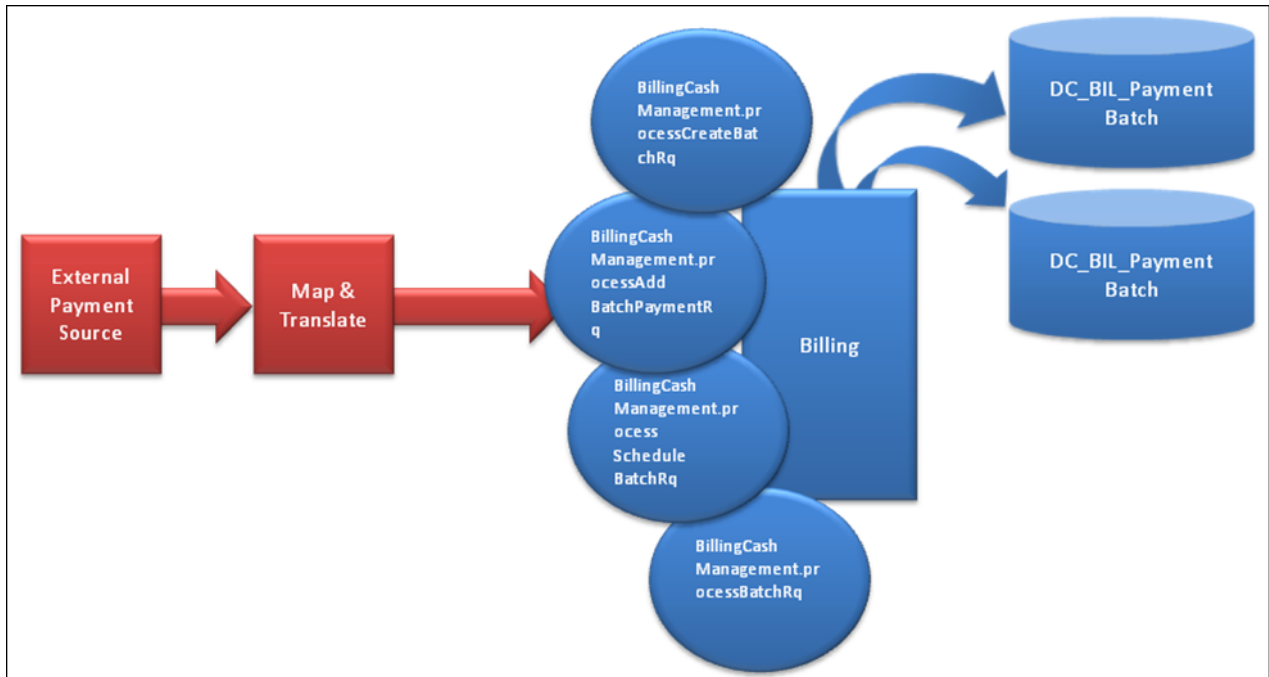
The amount entered cannot exceed the Available Amount.

The Items In Suspense page appears with the Items Pending Disbursement list displayed.

1. From the Account Home page, click **Items in Suspense**.
2. Select the check box for the appropriate item in suspense.
3. In the **Select An Action** list, click **Disburse**.
4. Select one of the following check boxes:
 - Refund To Check
 - Refund To EFT
5. Click **Submit**.
6. (Optional) Click **Select Another Payee** to name a different payee for the disbursement.
7. Select the appropriate party.
8. (Optional) In the **Disbursement Amount** text box, enter an amount other than the default to disburse less than the full amount.
9. Click **Refund**.
10. (Optional) Click the **View Details** link to see the details of the item pending disbursement.

Working with External Payment Processing

The integration between Duck Creek Billing™ and external payment processing/lock box systems is illustrated below.



Typically, a lock box is a post office box that is accessible by a bank or payment processor. The bank or processor retrieves the payments, processes them, and deposits them directly into the company's account. Payments are converted to electronic data, and the Duck Creek Billing™ lock box interface parses the incoming payment files using XSL extensions. The Duck Creek Billing™ Lock Box Interface handles the processing of incoming lock box files by creating and submitting payment batches into the system. The actual implementation is configurable by XSL templates used to map the lock box file to internal Duck Creek Technologies, Inc. Server requests.

The Duck Creek Billing Watch Directory Service monitors lock box directories and submits corresponding EXAMPLE Server® requests based on incoming files. A customer can implement a new lock box format by altering the XSL, or overriding the request with a custom implementation.

The lock box interface uses the following EXAMPLE Server® requests:

- <PaymentProcessing.processPaymentsRq> request
- <PaymentProcessing.testProcessPaymentsXSL> request

The following table includes the attributes of the <PaymentProcessing.processPaymentsRq> request.

Request Name	Request Name	Type	Required	Description
<PaymentProcessing.processPaymentsRq>	lockboxFile	String	Yes	Specifies the lockbox file to execute. The file is listed in the BillingInterfaces.xml configuration file.

When EXAMPLE Server® executes the <PaymentProcessing.processPaymentsRq> request, the PaymentProcessing object does the following:

- Checks the configuration file for the lock box information.
- Scans all the files in the specified lock box watch directory.
- Sorts each file into batches as specified in the file mapping.
- Executes the <BillingCashManagement.processCreateBatchRq> request for each batch.
- Executes the <BillingCashManagement.processAddBatchPaymentRq> request for each payment record.
- Executes the <BillingCashManagement.processBatchRq> request to process each batch after loading all payments.
- Copies all lock box files to the backup directory.
- Copies the file to the error directory and logs a process exception.

This <PaymentProcessing.testProcessPaymentsXSL> request was designed to help debug configuring the XSL files used to map the lock box file format to the XML request format used.

The request returns the XML output for each batch and payment record for a specified lock box file in the watch directory.

Configuring the Lockbox

To configure the lockbox interface:

1. Open the **Edit Configuration** utility.
2. In the Virtual Directory list, click the **Express** virtual directory.
3. Click the **Master Catalog** tab.
4. Browse to the **BillingInterfaces** section.
5. Set the **url** and **rootdir** attributes to the location of the BillingInterfaces.xml file.



Typically the location is C:\Program Files\Duck Creek Technologies\Shared\.

6. Click **Refresh**.
7. Click the **Billing Interfaces** tab.
8. Configure the lock box interface by adding or editing the appropriate lock box elements in the following table.

Attribute	Type	Required	Description
name	String	Yes	Specifies the name of the lock box instance.
path	String	Yes	Specifies the watch directory for the lock box instance.
currencyCulture	String	No	Specifies the currencyCulture for the lock box instance.

batchStartPosition	Int	Yes	Specifies the start position of the batch number in the lock box file mapping.
batchLength	Int	Yes	Specifies the length of the batch number in the lock box file mapping.
xsl/path	String	Yes	Specifies the directory for the XSL files.  The default path is C:\Program Files\Duck Creek Technologies\transform\
backup/path	String	Yes	Specifies the backup directory for the lock box files.
error/path	String	Yes	Specifies the error directory for the lock box files.
processCreateBatchRqXsl	String	Yes	Specifies the name of the XSL file to map
<BillingCashManagement.processAddBatchPaymentRq> request	String	Yes	Specifies the name of the XSL file to map
processScheduleBatchRqXsl	String	No	Specifies the name of the XSL file to map
processBatchRqXsl	String	Yes	Specifies the name of the XSL file to map

XSL Configuration

One of two methods can be used to customize the file mappings and the Billing Cash Management requests:

- Billing Cash Management API calls using the XSL extensions (for example, <xsl:value-of select="LockBoxBatch:getBatchNumber()"/>)
- Constant values or hard coded XSL text values (for example, <EntryUserId>Admin</EntryUserId>)

The following table contains the XSL extensions that can be used with API calls.

Name	Type	Use	Description
------	------	-----	-------------

<xsl:stylesheet xmlns:LockBoxBatch="urn:LockBoxBatch"/>	String	Required	Specifies the XSL extension prefix required to map to the LockBoxBatch object.
<xsl:value-of select="LockBoxBatch:getBatchNumber()"/>	String	Required	Gets the string for the batch number specified in the lock box configuration by LockBox/@currencyCulture & LockBox/@batchLength
<xsl:value-of select="LockBoxBatch:getProcessedDate()"/>	String	Required	Gets the current processing date
<xsl:value-of select="LockBoxBatch:getBatchId()"/>	Integer	Required	Maps the BatchId created in createBatchRq to subsequent requests (batchRq)
<xsl:stylesheet xmlns:LockBoxRecord="urn:LockBoxRecord"/>	Integer	Required	XSL extension prefix required to map to the LockBoxRecord object
<xsl:value-of select="LockBoxRecord:getSubString(,#, #)"/>	String	Required	Gets a substring from the lock box directory format given the string start position and length. (Extends .NET substring() function)
<xsl:value-of select="LockBoxRecord:getSubStringDecimal(,#, #)"/>	Decimal	Optional	Returns the same value as getSubString() but formats the return value as a decimal
<xsl:value-of select="LockBoxRecord:getBatchId()"/>	String	Required	Maps the BatchId created in createBatchRq to LockBoxRecords
<xsl:value-of select="LockBoxRecord:getProcessedDate()"/>	String	Required	Gets the current processing date

Example 1: XSL for the <BillingCashManagement.processCreateBatchRq> request

```

<xsl:template match="@* | node()">
  <BillingCashManagement.processCreateBatchRq>
    <PaymentBatchRecord>
      <BatchNumber>
        <xsl:value-of select="LockBoxBatch:getBatchNumber()"/>
      </BatchNumber>
      <ControlTotal>0</ControlTotal>
      <Description>
        LockBox Transmission <xsl:value-of select="LockBoxBatch:getProcessedDate()"/>
      </Description>
      <EntryUserId>Admin</EntryUserId>
      <CurrencyCulture>en-US</CurrencyCulture>
    </PaymentBatchRecord>
  </BillingCashManagement.processCreateBatchRq>
</template>

```

```

    </BillingCashManagement.processCreateBatchRq>
</xsl:template>

```

Example 2: XSL for the <BillingCashManagement.processAddBatchPaymentRq> Request

```

<xsl:template match="@* | node()">
  <BillingCashManagement.processAddBatchPaymentRq>
    <PaymentBatchEntryRecord>
      <PaymentBatchEntryId></PaymentBatchEntryId>
      <AccountId>
        </AccountId>
      <AccountReference>
        <xsl:value-of select="LockBoxRecord:getSubString(2,14)"/>
      </AccountReference>
      <PolicyReference>
        <xsl:value-of select="LockBoxRecord:getSubString(16,14)"/>
      </PolicyReference>
      <PaymentBatchId>
        <xsl:value-of select="LockBoxRecord:getBatchId()"/>
      </PaymentBatchId>
      <PaymentReference>
        </PaymentReference>
      <Amount>
        <xsl:value-of select="LockBoxRecord:getSubStringDecimal(36,10)"/>
      </Amount>
      <EntryUserId>Admin</EntryUserId>
      <CurrencyCulture>en-US</CurrencyCulture>
      <PaymentMethodCode>CK</PaymentMethodCode>
      <AllocationClassCode>P</AllocationClassCode>
      <PostMarkDate></PostMarkDate>
      <EntryDate>
        <xsl:value-of select="LockBoxRecord:getProcessedDate()"/>
      </EntryDate>
      <PaymentBatchEntryLockingTS>
        </PaymentBatchEntryLockingTS>
      <ProcessedTimestamp>
        </ProcessedTimestamp>
      <PaymentMethodDetail>
        </PaymentMethodDetail>
      <UnidentifiedRecord>
        </UnidentifiedRecord>
      </PaymentBatchEntryRecord>
    </BillingCashManagement.processAddBatchPaymentRq>
  </xsl:template>

```

Example 3: XSL for the <BillingCashManagement.processBatchRq> Request

```

<xsl:template match="@* | node()">
  <BillingCashManagement.processBatchRq>
    <PaymentBatchRecord>
      <PaymentBatch>
        <PaymentBatchId>
          <xsl:value-of select="LockBoxBatch:getBatchId()"/>
        </PaymentBatchId>

```

```

        </PaymentBatch>
    </PaymentBatchRecord>
</BillingCashManagement.processBatchRq>
</xsl:template>

```

Using the Watch Directory Service

The Duck Creek Watch Directory Service is designed to execute requests for inbound lock box files. The file hits the watch directory and then executes the corresponding lock box request with the lock box name.

To configure the DCT Watch Directory Service:

1. Browse to the **DuckCreek.Utility.WatchDirectorySvc.exe.config** located in the existing Shared directory at C:\Program Files\Duck Creek Technologies\ Shared\Bin.
2. Configure the Watch Directory Service using the required configuration settings contained in the following table.

Name	Type	Description
ServerConfig	String	Specifies the location of the Server.config file.
RequestUrl	String	Specifies the URL of the EXAMPLE Server® to execute requests.
DuckCreekUserID	String	Specifies the User Id to execute the requests.
DuckCreekPassword	String	Specifies the User password.

Following is an example of the configuration xml:

```

<configuration>
  <appSettings>
    <add key="InterfacesConfigPath" value="C:\Program Files\Duck Creek Technologies
\Shared\BillingInterfaces.xml"/>
  </appSettings>
</configuration>

```

Configuring the Watch Directory

After the lockbox configuration is complete, each watch directory must be created.

To configure a watch directory:

1. Open the **Edit Configuration** utility.
2. Click the **BillingInterfaces** tab.
3. Browse to **FileWatcherService**.
4. In the **path** attribute, specify the path to the lock box watch directory.



This is the same as the LockBoxInterface/LockBoxFiles/LockBox/@path value.

5. In the **action** attribute, enter the **<PaymentProcessing.processPaymentsRq>** request command with the **lockboxFile** attribute set to the name of the lock box instance.



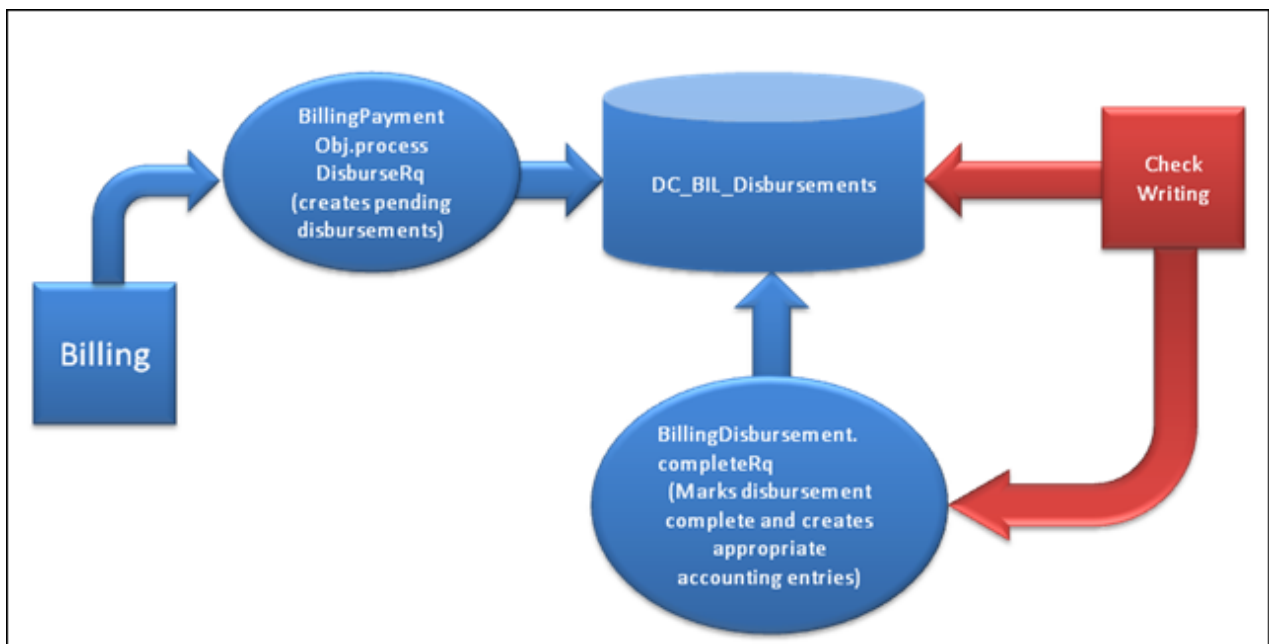
This is the same as the LockBoxInterface/LockBoxFiles/LockBox/@name.

Following is an example of the Watch Directory Service xml:

```
<Interfaces>
  <FileWatcherService>
    <WatchDirectory path="C:\Platform\Lockboxfiles\Bank1"
    action="PaymentProcessing.processPaymentsRq lockboxFile=&quot;Bank1&quot; " />
    <WatchDirectory path="C:\Platform\Lockboxfiles\*"
    action="PaymentProcessing.processPaymentsRq lockboxFile=&quot;""&quot; " />
  </FileWatcherService>
</Interfaces>
```

Working with the Disbursement Service Interface

The Disbursement Service Interface integrates Duck Creek Billing™ with external check writing/disbursement systems. It is accessible to users from the Billing Administration tab in Duck Creek Billing™, but must be configured for use.



Working with the Disbursement Interface Service

The Disbursement Interface Service uses the EXAMPLE Scheduled Request Queue to run a server request and process pending disbursements that have been requested in Duck Creek Billing™. The server request processes a pre-defined quantity of disbursements, and then reschedules itself for processing future disbursements based on the configuration settings specified.

The Disbursement Service Interface does the following:

- EXAMPLE Server® writes disbursements to the database when a user disburses money from EXAMPLE Express®
- Requests are scheduled
- The Scheduled Activity Service calls the <BillingDisbursementInterface.processDisbursementInterfaceRq> request for disbursements
- Schedules the first request through use of the Disbursement Interface on the Billing Administration tab in Duck Creek Billing™
- Reads pending disbursements and processes them using a customized request
- Calls the <BillingDisbursement.completeRq> request manually to complete disbursements
- The Disbursement Interface Service writes the disbursement detail records in the appropriate format (xml, flat file or web service), and changes the disbursement status to SB (submitted).

The Disbursement Interface Service implementation is configurable by overriding the DCT <BillingDisbursementInterface.processDisbursementInterfaceRq> request with a custom request.

Configuring the Disbursement Interface Service

Configuration settings must be specified to control the function of the disbursement interface.



By default, the Disbursement Interface will return an error unless the disbursement interface has been written.

The following table lists the configuration settings for the disbursement interface, and these settings are typically overridden by a custom request.

Configuration Setting	Definition	Default
StartTime	Specifies the earliest time in the day the interface will check for disbursements.	08:00:00 An empty node means no restriction.
InterfaceWindow	Specifies the number of minutes from the start time that requests can be processed. Using duration rather than a specific time makes it easier to process through midnight.	600
Frequency	Specifies in seconds how often the interface will check for unprocessed pending disbursements.	600
RecordsPerCycle	Specifies the maximum number of disbursements that will be processed per cycle.	100 An empty node or value of 0 means unlimited disbursements processed per cycle.

For information about using the disbursement interface, see *Working in the Billing Administration Area* in the *Duck Creek Billing™ User Guide*. For more information about customization and integration, contact your Account Manager.

Configuring the Disbursement Authorization Privilege

The Disbursement Authorization feature allows Duck Creek Billing™ users to create a threshold for automatic disbursements, and requires a specific privilege to enable a user to authorize a disbursement.

Disbursements can be processed automatically without any authorization. Out-of-the-box, the account authorization limit default is 0, which prevents automatic disbursements.



For more information about using this feature, see *Working with Disbursement Authorization* in the *Duck Creek Billing™ User Guide*.

Specifying the Automatic Disbursement Value

To allow automatic disbursements up to a specific amount:

- Change the value of the *AutomaticDisbursementAuthorizationLimit* field in the *Disbursement_AccountPlan* group in the *Billing_Policy_Account_Common_1_0_0_0* Manuscript to the appropriate value.

Automatic disbursements will now be processed up to and including this amount with no authorization required.

Working with the Disbursement Authorization Privilege

To create the required privilege for authorizing disbursements and associate that privilege with a user:



Only one disbursement authorization limit may be designated for a user.

1. Open EXAMPLE UserAdmin™, and in the left navigation panel, click **Add a role**.
2. Enter a role name and description in the appropriate text boxes.
3. Click **Select General System** and then select the appropriate privileges.
4. Click **Select Policy Administration** and then select the appropriate privileges.
5. Click **Billing Administration** and then select the appropriate privileges.
6. Click **Author Administration**, select the appropriate privileges, and the click **Finish**.
7. Click the **View Entities Associated with this role** button  for the newly created role.
8. Click **Add more entities to role**.
9. Enter a name in the **Enter Name** text box and then click **Search**.
10. Select the appropriate entities, and then click **Add roles to these entities**.
11. Click **Manage role/privilege associations** in the left navigation panel.
12. In the **Role** list click the appropriate role, and then in the **Privilege** list, click **BIL_AuthorizeDisbursement**.
13. Enter **DisbursementAuthorizationLimit** in the **Name** text box, enter the appropriate value in the **Value** text box, and then click **Insert**.
14. To create a new user or associate this role with an existing user, see *Working with Users*.

The Disbursements Pending Authorization report is accessible from the Billing Dashboard, and lists all pending disbursements awaiting authorization. This report can be used to verify and authorize disbursements for Duck Creek Billing™ accounts.

The Disbursement Activity report, also accessible from the Billing Dashboard, lists all the processed disbursements.

The Disbursement Interface Service processes all Duck Creek Billing™ disbursements whether they originate from within Duck Creek Billing™ or are paid using a system that is external to Duck Creek Billing™. External disbursements are subject to the same authorization rules as internal disbursements.



A disbursement cannot be requested and authorized by the same Duck Creek Billing™ user.

Configuring the Disbursement Interface Service

Configuration settings must be specified to control the function of the disbursement interface.



By default, the Disbursement Interface will return an error unless the disbursement interface has been written.

The following table lists the configuration settings for the disbursement interface, and these settings are typically overridden by a custom request.

Configuration Setting	Definition	Default
StartTime	Specifies the earliest time in the day the interface will check for disbursements.	08:00:00 An empty node means no restriction.
InterfaceWindow	Specifies the number of minutes from the start time that requests can be processed. Using duration rather than a specific time makes it easier to process through midnight.	600
Frequency	Specifies in seconds how often the interface will check for unprocessed pending disbursements.	600
RecordsPerCycle	Specifies the maximum number of disbursements that will be processed per cycle.	100 An empty node or value of 0 means unlimited disbursements processed per cycle.

For information about using the disbursement interface, see *Working in the Billing Administration Area* in the *Duck Creek Billing™ User Guide*. For more information about customization and integration, contact your Account Manager.

The Disbursement Interface

The following structure details the data available from the standard Duck Creek Billing Disbursement Interface.

```
<BillingDisbursement.getDisbursementDetailRs status="success">
  <Disbursement>
    <AccountId>1</AccountId>
    <DisbursementId>1</DisbursementId>
    <DisbursementRequestDate>2009-10-06</DisbursementRequestDate>
    <DisbursalDate xsi:nil="true"/>
  </Disbursement>
</BillingDisbursement.getDisbursementDetailRs>
```

```

<DisbursementUserFullName>Express Admin</DisbursementUserFullName>
<DisbursementMethodCode>CK</DisbursementMethodCode>
<DisbursementMethod>Check</DisbursementMethod>
<DisbursementStatusCode>R</DisbursementStatusCode>
<DisbursementStatus>Requested</DisbursementStatus>
<ReversalReason xsi:nil="true"/>
<ReversalDate xsi:nil="true"/>
<RebillReason xsi:nil="true"/>
<RebillDate xsi:nil="true"/>
<CurrencyCulture>en-US</CurrencyCulture>
<Amount>1.9400</Amount>
<ProcessingOrgUnitCode>1</ProcessingOrgUnitCode>
<DisbursementLockingTS>AAAAAAhDQ8=</DisbursementLockingTS>
<AssociatedParties>
  <Party PartyId="10" Role="Payee"/>
</AssociatedParties>
<Parties>
  <PartyRecord>
    <PartyRecordPartyId>10</PartyRecordPartyId>
    <Party>
      <PartyName>Name</PartyName>
      <PartyFirstName>Test</PartyFirstName>
      <PartyMiddleName xsi:nil="true"/>
      <PartyNamePrefix xsi:nil="true"/>
      <PartyNameSuffix xsi:nil="true"/>
      <PartyGender xsi:nil="true"/>
      <PartyBirthDate xsi:nil="true"/>
      <PartyCorrespondenceName>Test Name</PartyCorrespondenceName>
      <PartyTypeCode>P</PartyTypeCode>
    </Party>
    <PartyPhones>
      <PartyPhone>
        <PhoneNumber>8031234567</PhoneNumber>
        <PhoneExtension xsi:nil="true"/>
        <PhoneCountryCode xsi:nil="true"/>
        <PhoneFormatCode xsi:nil="true"/>
      </PartyPhone>
    </PartyPhones>
    <PartyEmails>
      <PartyEmail>
        <EmailAddress>testname@xyz.com</EmailAddress>
      </PartyEmail>
    </PartyEmails>
    <Locations>

```



```

        <Location>
            <LocationAddressLine1>Address 1</LocationAddressLine1>
            <LocationAddressLine2 xsi:nil="true"/>
            <LocationCounty xsi:nil="true"/>
            <LocationCity>City</LocationCity>
            <LocationStateCode>MO</LocationStateCode>
            <LocationPostalCode>77777</LocationPostalCode>
            <LocationCountryCode xsi:nil="true"/>
        </Location>
    </Locations>
</PartyRecord>
</Parties>
</Disbursement>
</BillingDisbursement.getDisbursementDetailRs>

```

Working with Reports and Letters

Working with Reports

The following are requirements for working with reports in Duck Creek Billing™:

- The out-of-the-box reports must be accessible in SQL Server Reporting Services (SSRS)
- SSRS must be integrated with Duck Creek Billing™
- A user must be assigned the appropriate privileges to view reports

Verifying Report Accessibility

Before integrating reports with Duck Creek Billing™, you must verify the out-of-the-box reports are deployed and accessible in SQL Services Reporting Services. Even after integration, reports will not work in Duck Creek Billing™ if they do not work in SQL Services Reporting Services prior to integration.

To verify the reports are deployed and accessible:

1. Navigate to the **Reports Server**.
2. Open the base folder containing reports, and navigate to the folder containing the daily reports installed out-of-the-box with Duck Creek Billing™.



Make note of the base folder name and the URL for the Reports Server. This information is needed to complete integration with Duck Creek Billing™.

3. Open one of the reports and verify it works as expected.
4. Close SQL Server Reporting Services.

Integrating EXAMPLE Server® and SQL Server Reporting Services

Regardless of when the services are started, integration with EXAMPLE Server® is required to use the reporting features in Duck Creek Billing™.

To integrate SQL Server Reporting Services with EXAMPLE Server®:

1. Open the **Edit Configuration** utility.



The default location for this utility is C:\Program Files\Duck Creek Technologies\Util\EditConfig.exe.

2. In the **Virtual directory** list, click the EXAMPLE Server® virtual directory.
3. Click the **Server.Config** tab.
4. Navigate to the **Reporting** section.
5. Set the **ReportServerInterfaceEnabled** setting to **True**.
6. Set the **ReportServerUrl** to the URL for the virtual directory for your report server's Web service. For example, `http://<report server name>ReportServer`.
7. Set the value of the **BaseFolder** setting to the base folder containing the Duck Creek Billing™ reports.
8. Click **Save**, **Bounce**, and then click **Close**.
9. Navigate to the **ReportServerUrl** specified in Step 6.
10. Click the name of the base folder specified in Step 7.
11. Click **Data Sources**, and then click **Example Data**.
12. Click the **Properties** tab, and then click **General** in the left navigation panel.
13. Select **Windows Integrated Security**.
14. Click **Apply**, and then close the browser.

Assigning Report Access Privileges

A user must have the appropriate EXAMPLE UserAdmin™ privileges to view or access reports.



A user can be granted access to view all available reports or access can be granted for specific reports.

To ensure the required privileges are assigned:

1. Navigate to **EXAMPLE UserAdmin™**.
2. Select the appropriate user, and then view the privileges associated with the user.
3. Select the appropriate report privileges, and then click **Save** and **Close**.

The reports available on the Billing Dashboard are determined by the assigned privileges of the logged in user.

Creating a Custom Report

You can create custom Duck Creek Billing™ reports using Microsoft SQL Server Business Intelligence Development Studio.

Consider the following guidelines for any custom report created for use with Duck Creek Billing™:

- Reports must include the following parameters:
 - **ProcessingDate**, to specify the run date the report
 - **CultureCode**, to specify the culture
- Although reports may be viewed primarily online, custom reports should be tested in different formats in SQL Server Reporting Services to ensure they render as expected.

- The queries the reports are based on should be highly optimized for performance because many reports run against the transactional database.

To create a custom report:

1. Create the report using Microsoft SQL Server Business Intelligence Development Studio.
2. Save the report.



Duck Creek Technologies, Inc. recommends saving the report to the base folder where the out-of-the-box Duck Creek Billing™ reports are located.

3. Verify that the custom report can be opened using SQL Server Reporting Services.
4. Run SQL Server Management Studio and connect to the Example Data database.
5. Execute the following SQL statement:

```
INSERT INTO DC_PLT_Reports (Name, FullPath)
VALUES ('Display Name', 'Report path');
```
6. Change the **Display Name** to the report name you wish to display on the Billing Dashboard.
7. To change the **Report Path** to the path of the report within SQL Server Reporting Services, perform one of the following actions:
 - If the report is located in the base folder, specify a relative path (for example, `./Billing/Monthly/Aged Receivables`).
 - If the report is not located in the base folder but is still on the ReportServerUrl, specify the report path, starting at the root folder level (for example, `/Example Server/Billing/Monthly/Aged Receivables`).
 - If the report is located on a server other than ReportServerUrl, specify the full URL (for example, `http://localhost/ReportServer/?/EXAMPLE Server/Billing/Monthly/Aged Receivables`).

The custom report appears in the Reports list on the Billing Dashboard.

Out-of-the-Box Reports

The following table lists the out-of-the-box reports installed with Duck Creek Billing™.

Report Name	Description
Aged Receivables	Lists all aged receivables.
Cash in Suspense	Lists all cash in suspense.
Daily Cash Journal	The daily cash journal.

Unidentified Cash	Lists unidentified cash entered into the system.
Commission Activity	Displays all authorized commissions that occurred on a particular day. This report may be used to interface with a customer's check-writing system that actually pays commissions to agents. The interface may be configured at implementation time. The system will extract information from the Bill Item and Commission Authorization tables during the daily batch process. It then produces a daily report of all commissions that have been authorized that day.
Disbursement Activity	Includes all disbursement activity for Duck Creek Billing™ accounts.
Disbursements Pending Authorization	Lists all pending disbursements awaiting disbursement authorization.
General Ledger	Displays the amounts on the general ledger to date. This is not a daily scheduled report but is produced when requested.
Payment Batch Activity	Detailed account of payment batch activity.
Transaction Balancing	Queries all of the database tables to ensure everything is in balance.

Working with Manual Letter Generation

Duck Creek Billing™ functionality provides a framework for customization to allow the user configure a menu link in the left navigation panel of a specified page to open a particular Manuscript page. Examples of this functionality that are available out of the box include the EFT Maintenance page and the Commission Plans page.

Using this framework, after customization, users can manually trigger the generation of specified letters from Duck Creek Billing™ for viewing and printing. The letters are generated by clicking the **Account Workflow** link in the left navigation panel from the Account Home page for the appropriate account. Letters can also be generated by clicking the **Policy Workflow** link on the left navigation panel of the Policy Term Summary page for a specific policy associated with that account.



The link names may be customized by the user during implementation.

When a letter is printed, that activity appears on the **Recent Account Activity** list for the account, and includes a paperclip link  that allows the user to open the letter for viewing.

To print a letter using Manual Letter Generation:

1. From the left navigation panel on the Account Home page, click **Account Workflow**.



You can also:

From the left navigation panel on the Policy Summary page, click **Policy Workflow**.

A list of available letters appears.

2. Click the appropriate item in the list.
3. Click **Print** to print the selected letter, or click **Cancel** to return to the Account Home page.
For information about this feature contact your Duck Creek Technologies Account Representative.

Working with Automatic Letter Generation

When implemented, the Automatic Letter Generation feature allows users to integrate a third-party forms provider to manage letters that are generated automatically.

Out of the box, generated letters (in the form of XML) are placed in the following directory: C:\Program Files\Duck Creek Technologies\External Documents.

The appropriate letter is generated automatically when one of the following events occur:

- Billing_PostPaymentReversalEvent
- Billing_EFT_RepresentEvent
- Billing_ReversePaymentEvent
- Billing_EFT_RejectEvent
- Billing_PostDisbursementEvent
- Billing_EFT_ChangeEvent

Each event triggers a specific ManuScript which automatically generates the appropriate letter for the event. The form data is generated as XML, and can be sent to a third-party forms provider. The conditions that trigger the events can be configured or specified to provide additional flexibility. For example, the system could automatically create an EFT payment rejection letter the same day the payment is reversed in the billing system. This provides the benefit of prompt notification to the payor regarding the status of their payment, and provides information regarding the action required to prevent cancellation of the associated policy or policies.

Other letters that could be automatically generated include notification of:

- Representment of an EFT payment following an initial rejection.
- EFT or other payment type rejection.
- Disbursement from the account.
- Changes to the EFT information held for the account.

When a letter is automatically generated for an account, the **Recent Account Activity** list is updated to include the form name, the name of the third-party provider, a date and time stamp, a success or failure message, and link that opens the letter for viewing.

The DCT_DocumentInterface section in the Server.config file contains the parameters used by the component map and must be customized for each carrier's needs.

Enabling the Automated Letter Generation Feature

To use this feature following ManuScript customization:

1. Navigate to the Edit Configuration Utility (located at Global Root Directory\Util\EditConfig.exe), select the appropriate Virtual Directory, and open the Server.config file.
2. In the ExternalDocumentsFolderRootPath section, locate the UseExternalDocumentInterface section and change the default value of **False** to **True**.
3. Click **Save**, and then click **Bounce**.
The Automated Letter Generation feature is now enabled.

For more information, contact your Account Manager.

Troubleshooting

Troubleshooting Scheduled Activities

Duck Creek Billing™ uses two services for scheduled activities processing:

- The Duck Creek Scheduled Requests EXAMPLE Billing Components 0
- The DuckCreek_Async_Request_Queue_ EXAMPLE AsyncQueue 0



If scheduled activities do not work as expected, verify that the Duck Creek Scheduled Requests EXAMPLE Billing Components 0 and the DuckCreek_Async_Request_Queue_ EXAMPLE AsyncQueue 0 services are running.

To verify the services are running:

1. From the **Start** menu, click **Control Panel**.
2. Click **Administrative Tools**.
3. Click **Services**.
4. Locate and click **Duck Creek Scheduled Requests EXAMPLE Billing Components 0**.
5. Do one of the following:
 - Click **Start**.
 - Click **Restart**.
6. Locate and click **DuckCreek_Async_Request_Queue_EXAMPLE AsyncQueue 0**.
7. Repeat Step 5 .

Following are the steps involved in scheduled activities processing:

1. The Duck Creek Scheduled Requests EXAMPLE Billing Components 0 service queries for outstanding scheduled activities in the DC_BIL_ScheduledActivities table.
2. Any outstanding activities are set to process, and a request to process the appropriate activities is sent to the server.
3. The request is processed by the server, and asynchronous activity is scheduled if required.
4. Any scheduled asynchronous activity is sent to the Semaphore table.

5. The DuckCreek_Async_Request_Queue_EXAMPLE AsyncQueue 0 reads the Semaphore table and executes the request.
6. After the request is completed, the scheduled activity item is set to a status of C (Complete) in the DC_BIL_ScheduledActivities table.

If the scheduled activities fail to process, try the following troubleshooting tips.

Checking Configuration Files

Verify that the settings in the DuckCreek.Utility.ScheduledRequestSvc.exe.config and the AsyncQueueService.exe.config file are correct. In the Duck Creek root directory (for example, C:\Program Files\Duck Creek Technologies\) navigate to the appropriate file.

The following table lists the configuration settings in the DuckCreek.Utility.ScheduledRequestSvc.exe.config file. To modify these settings, open the Edit Config Utility located by default at C:\Program Files\Duck Creek Technologies\Util\EditConfig.exe or another text editor program.

Key	Default Value	Definition
ServerConfig	C:\Program Files\Duck Creek Technologies\Shared\Server.Config	The location of the Server.config file.
RequestIntervalInMilliseconds	60000	Specifies the time in milliseconds between checking for requests.
DuckCreekUserID	admin	The user ID.
DuckCreekPassword	admin	The user password.
RequestUrl	<i>http://localhost:8080/DuckCreek/dctserver.aspx</i>	URL of the EXAMPLE Server® responsible for working with the specified web site to ensure the correct application pool is recycled.
AppLogSource	DCTSchedRequestSvc	The source description used by the service when logging an error to the Event Log.
AppLogName	DCTSchedRequestLog	The name of the Event Log to log errors for this service.
RequestFile	ScheduledRequests.xml	The XML containing the request.
RequestToCheckForSuccess	ScheduledActivity.queueListRq	The name of the request to check for successful activity scheduling.
RequestTimeoutInMilliseconds	120000	The amount of time in milliseconds the service waits for EXAMPLE Server® to respond when sending a request to the URL.

The following table lists the configuration settings in the AsyncQueueService.exe.config file. To modify these settings, open the Edit Config Utility that is located by default at C:\Program Files\Duck Creek Technologies\Util\EditConfig.exe.

Key	Default Value	Description
ServerConfig	C:\Program Files\Duck Creek Technologies\shared\Server.Config	The location of the Server.config file.
RequestIntervalInMilliseconds	60000	Specifies the time in milliseconds between checking for requests.

Working with the Windows Events Log

If the Windows Events log is full, it may cause the DuckCreek_Async_Request_Queue_EXAMPLE AsyncQueue 0 service to terminate without processing requests.

To troubleshoot the Windows Events log:

1. Click **Start**, and then click **Control Panel**.
2. Click **Performance and Maintenance**, and then click **Administrative Tools**.
3. Double-click **Computer Management**, and then expand the **Event Viewer**.
4. Click **Application** to view the logged events.
5. Review the event logs for errors.
6. Right-click **Application** to open the log file or save it to a different file.
7. Right-click **Application** to clear the appropriate events log.



All scheduled activities for a given day must be executed before subsequent activities will be processed.

The following table lists the activity status codes associated with items in the scheduled activity table.

Activity Status Code	State	Description
C	Complete	The scheduled activity is successfully executed.
F	Failed	The scheduled activity failed to execute.
H	Hold	Execution of the scheduled activity is on hold.
I	Processing	The scheduled activity is in the process of executing.
P	Pending	Execution of the scheduled activity is pending.
X	Expired	The scheduled activity has expired without executing.

Working with Asynchronous Activity

To verify asynchronous activity:



The table should be empty unless there are outstanding requests.

- Check the Semaphore table to see if it contains any outstanding requests.

Checking for Failed Scheduled Activities

The DC_BIL_ProcessException table stores exception information about failed requests. This table can be queried for any processing exceptions that may have occurred when a scheduled activity has failed. Any scheduled activities that have failed to process are logged in this table.

Re-launching a Scheduled Activity

If the cause of a scheduled activity processing failure is not known after following the troubleshooting tips, re-launching the system may result in successful execution of the scheduled activity.

To re-launch a scheduled activity:

The scheduled activity server will attempt to re-execute the request.

1. In the **DC_BIL_ScheduledActivities** table locate any activity with a code of **I** (Processing).
2. Change the code to **P** (Pending).
3. If this is not successful, set the code to **P** while monitoring with TraceMonitor to help locate the error. For more information about using TraceMonitor, see *EXAMPLE TraceMonitor* in *Utilities*.

Troubleshooting the MSMQ Billing Listener

The DuckCreek_MSMQBillingListener_EXAMPLE Billing Components 0 service logs the requests and responses for errors to the BillingListenerErrors file.



The requests and responses for errors logged by the MSMQ Billing Listener are automatically placed in the BillingListenerErrors file, by default, located in the Duck Creek root directory (for example, C:\Program Files\Duck Creek Technologies). This file is dynamically generated when errors are logged.

The request and response files for Duck Creek Billing™ errors share the same unique name. For example, if a request is identified as 8193.Rq.xml, the corresponding response would be identified as 8193.Rs.xml.

For more information see *Viewing the Error Queue* in the *Duck Creek Billing™ User Guide*.

Troubleshooting with the Configuration Inspector Utility

The Configuration Inspector utility, located in the EXAMPLE Utility Center, can be used to inspect Duck Creek Billing™ and identify issues with ManuScripts or configuration settings. For more information, see *Working with the Configuration Inspector* in the *EXAMPLE Utility Center*.